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ONCE A JOLU, ALWAYS A JOLU – PURE PRIDE

Alumni Bulletin Year I Vol II

JOLU BUZZ

SPECIAL EDITION

THE JALPAIGURI GOVERNMENT ENGINEERING COLLEGE ALUMNI ASSOCIATION

in loving memory



It is with profound respect and heartfelt gratitude that the Jalpaiguri Government Engineering College Alumni Association (JGECAA) dedicates this space to all our departed alumni. Their lives, continue to shape our world in countless ways. Their contributions, are woven into the very fabric of our shared history.

The Jalpaiguri Government Engineering College Alumni Association



FROM THE DESK OF THE MANAGING COMMITTEE OF THE JGEC ALUMNI ASSOCIATION

"Journey towards Horizon" the JGEC Alumni Association



President
Bhaskar Dasgupta
Mechanical, 1983



Secretary
Partha Haldar
Civil, 1993

From The President & Secretary of JGECAA

The journey of the JGEC Alumni Association began around 2010 during the NBA accreditation process of Jalpaiguri Government Engineering College. At that time, the college needed to secure good scores for accreditation, and it was realized by the then Principal, Dr. Joytirmoy Jhampati, that having an active alumni association within the college campus would significantly contribute to this objective.

Although a Kolkata-based alumni association already existed, its activities were centered outside the campus. Dr. Jhampati took the initiative to connect with alumni in Siliguri and Jalpaiguri and facilitated the establishment of a campus-based alumni body. The association began its journey from a small room within the college premises, provided by the Principal.

The JGEC Alumni Association formally started its activities under the leadership of President Dipak Sarkar (1st batch) and Secretary Goutam Bhaduri (1983 batch). The Principal also extended initial financial support of ₹5,000 to open a bank account at SBI, Danguajhar. With the collective effort of dedicated alumni, the association played a meaningful role in helping the college achieve NBA accreditation. A major milestone was achieved in 2013 when alumnus Premangsu Ghosh (1979 batch) generously donated a piece of land within the campus for the association. This led to the construction of the Annex Building, funded by contributions from alumni. The Annex Building was successfully completed in 2016 and we got his own destiny.



The same year marked the beginning of an ambitious project—Alumni House “Kanchanjungha”, initiated by Mr. Alope Basu, an alumnus and businessman based in Qatar. With continuous support from alumni, the ground floor of the Kanchanjungha Convention Centre was completed in 2020, and all alumni activities were shifted there. The ground floor houses the alumni office and a seminar hall, covering approximately 3,000 sq. ft.

Subsequently, the first floor—comprising six well-equipped guest rooms—was constructed and completed within a few years. This facility now accommodates visiting recruiters and alumni, replacing earlier dependence on hotels.

Parallel to infrastructure development, the association launched its scholarship program in 2013 with just four scholarships totaling ₹80,000, supported by donors like Premangsu Ghosh and Bibhas Bhowmik (1985 batch). Over the years, this initiative has grown significantly. By 2025, 76 students are receiving scholarships, supported by 32 donors, with a total disbursement of ₹14.35 lakhs annually. The selection process involves both online and face-to-face interactions conducted by donors and the scholarship committee.

During the COVID-19 pandemic, the association actively supported casual and mess staff by distributing rations multiple times. The Annex Building also served as a temporary residence for an affected faculty member.

The first floor of Kanchanjungha was inaugurated in the presence of former Principal Dr. Haripada Bhowmik, Dr. Jhampati, and many distinguished alumni. The association has also extended support to “Jyoti,” a school for underprivileged children run by college students, by providing books, stationery, and clothing.

In addition to welfare activities, the association regularly organizes inter-college innovation programs, seminars, and lectures by experts from industry and academia. Initiatives such as a modern library, internet-enabled facilities, and spoken English classes have been introduced to enhance student development.

Currently, construction of a state-of-the-art convention centre on the second floor of Kanchanjungha is underway. The estimated value of the Kanchanjungha building stands at approximately ₹1.12 crore, with furniture worth ₹4.5 lakhs. The association now boasts around 2,300 registered members with unique ID numbers.

With a strong foundation and growing alumni support, the JGEC Alumni Association continues to expand its activities and remains committed to the academic and professional advancement of students of JOLU .



Log of editors: Chaos, Coffee, Repeat... Reloaded

If the last edition was our nostalgic warm-up lap, this one is us stretching properly, cracking a few joints (mostly knees and backs), and getting back into the long-distance run called “staying connected.”

JGEC, as we left it, was a place where time moved strangely. Four years felt like four minutes during adda, and four hours felt like four lifetimes during exams. And now, years later, time has done what it always does—moved on without asking permission. But here’s the curious thing: Jalu hasn’t really changed. Not where it matters.

The buildings may have acquired fresh coats of paint (or at least intentions), the labs may have upgraded from “experimental” to “semi-functional,” and students now debug more code than circuits—but the spirit? Still gloriously intact. Chaos, ingenuity, jugaad, and a stubborn refusal to give up—that’s still the curriculum no syllabus ever mentioned.

We often joke that JGEC didn’t prepare us for the real world. That’s not entirely true. It prepared us perfectly—just not in ways HR departments can quantify. It taught us how to survive on limited resources, negotiate impossible deadlines, and work in teams where “coordination” meant shouting across rooms. In hindsight, that’s not education—that’s elite training.

And speaking of training, the Alumni Association continues to evolve—not as a ceremonial body, but as a working ecosystem. Over the past months, what began as scattered enthusiasm is slowly shaping into something structured and meaningful. The focus is clear: not just reminiscing about the past, but actively contributing to the present and future.

Mentorship programs are gaining traction. Conversations that once happened over tea are now happening over calls and meets—with the same sincerity, just fewer samosas (a matter we must urgently address). Support systems are being strengthened, and initiatives are moving from “good ideas” to “actual execution”—a rare and commendable transition.

Let’s be honest—most alumni groups fade into two things: forwarded messages and annual nostalgia spikes. We are trying, quite deliberately, to avoid that fate. This bulletin itself is proof that consistency is being taken seriously. And consistency, as any old Jaluite will tell you, is harder than passing Applied Mechanics on the first attempt.

Now, a quick reality check with a pinch of humor.

We used to think the toughest questions in life were set by professors. Turns out, life has a much larger question bank—and no supplementary exams. Back then, we worried about attendance percentages. Today, we worry about cholesterol percentages. Progress, of a sort.

But in all this, what remains invaluable is connection. Not the digital kind—we have plenty of that—but the human kind. The kind where a senior picks up the phone and says, “Don’t worry, we’ll figure it out.” The kind where a batchmate you haven’t spoken to in 20 years still remembers your nickname but not your official name.

That’s the network we are building. Quietly, steadily.

This edition continues our effort to strike a balance—between memory and movement. You’ll find stories that make you smile, updates that keep you informed, and perhaps a few reminders that you are still part of something larger than your current designation or city.

We also want to acknowledge something important: participation. Many of you have written in, shared anecdotes, photos, and suggestions. Some of you have even volunteered time and effort. That matters. Institutions don't run on infrastructure alone—they run on involvement.

So, keep it coming. Send us your stories—the polished ones and the embarrassing ones (especially the embarrassing ones). Share your journeys, your lessons, your failures that turned into successes (or at least good stories over tea).

And for those still hesitating—consider this your gentle nudge. Or a firm push, depending on your personality.

Before we close, a small reflection.

Back in college, we often looked ahead—waiting to “get out,” to “become something.” Now, years later, we find ourselves looking back—not out of regret, but with a certain clarity. We didn't just become engineers there. We became resilient, adaptable, and just a little bit eccentric.

Not a bad deal, in hindsight.

So here's to the next chapter—of this bulletin, of the Association, and of this ever-expanding Jalu network.

Long live the spirit.

And may your deadlines be flexible, your networks stable, and your memories always slightly exaggerated for better storytelling.

For and on behalf of the JGECAA bulletin vertical
(Still running on Chaos, Coffee, and a remarkable lack of sleep)



Partha Pratim Ghosh
JGEC 1991



Sudip Mandal
JGEC 1992



Mitul Ranjan Chakraborty
JGEC 1996



Debendra Diasi
JGEC 2021

KANCHENJUNGA: IT'S ALL ABOUT DREAM AND REALIZATION OF THE DREAM



Nirjhar Chakravorti
Mechanical 2000

Nirjhar Chakravorti is a 2000-batch alumnus and an author of several books on organizational and sustainability transformation, published by Taylor and Francis (Routledge), New York.



Rajib Bhattacharya
Mechanical 2000

Rajib Bhattacharya is also a 2000-batch alumnus and a seasoned project management professional, who has played a pivotal role in driving and executing the Kanchenjunga construction initiative.

The journey of the Jalpaiguri Government Engineering College Alumni Association is not merely a story of organizational growth—it is a story of belonging, gratitude, and an enduring bond with one's alma mater.

Nearly fifteen to sixteen years ago, within the familiar grounds of the Jalpaiguri Government Engineering College campus, a group of committed alumni came together with a shared conviction: that the relationship between alumni and their college should not fade with time, but instead grow stronger with purpose. Their vision was simple yet profound—to remain connected to the campus, to engage with students, and to contribute meaningfully to the institution that shaped their lives.

From the very beginning, the Alumni Association immersed itself in campus life. Seminars and workshops were organized to broaden student horizons, collaborations were forged with external organizations, and scholarships were instituted to support young minds with promise but limited means. With each initiative, the Association strengthened its presence and reaffirmed its commitment. To reach a wider audience and to provide structure to its growing activities, an official website was also launched—marking another step in its evolving journey.

As the years passed and activities expanded, a deeper realization emerged. For these initiatives to truly endure, and for alumni engagement to be sustained across generations, the Association needed more than intent—it needed own infrastructure. A permanent space within the campus was envisioned as a place where memories, mentorship, learning, and service could come together under one roof.

The Alumni Association approached the college authorities with this aspiration. Yet, as is often the case with government institutions, procedural and administrative limitations made the allocation of land within the campus a formidable challenge. Still, hope remained alive, and alternative paths were quietly explored.

Then came a day that would change everything.

It was the last weekend of July 2013. On that day, when the campus hosted an alumni-organized seminar and witnessed the awarding of the Association's very first scholarships, the Annual General Meeting was held in the college conference room.

Amid discussions, reflections, and collective resolve, destiny intervened. The governing body learned of a small piece of privately owned land situated within the campus itself—an unlikely island of possibility. The landowners were willing to sell.

What followed was a moment of extraordinary unity and resolve. The Alumni Association reached out immediately, and on that very day, a mutual agreement was achieved.



Then, in a gesture that would forever be etched into the Association's history, a distinguished alumnus of the 1979 batch, Premangshu Ghosh, stepped forward.

Already a sponsor of one of the Association's earliest scholarships, Premangshu Ghosh, without hesitation or solicitation, offered to fully sponsor the purchase of the land. His generosity was not just financial—it was deeply symbolic. It reflected gratitude, faith, and an unwavering belief in giving back to the institution that had once nurtured his own dreams.

That day became a defining chapter in the story of the Jalpaiguri Government Engineering College Alumni Association. It transformed a long-held vision into a tangible foundation. It proved that when purpose meets compassion, history is made—not through grand declarations, but through selfless action.

What stands today is more than land or infrastructure. It is a testament to alumni spirit, collective memory, and one individual's extraordinary generosity.

The rest, indeed, is history—but a history that continues to inspire generations to come.

The closing months of the 2013 calendar year marked a turning point of historic significance for the Jalpaiguri Government Engineering College Alumni Association.

During the last quarter of 2013, after years of vision, perseverance, and collective effort, the Alumni Association formally registered the land in accordance with the law of the land. This moment completed a long and determined journey—one that transformed a deeply held aspiration into legal and institutional reality. The process was led by the Association's then Secretary, Dr. Amaresh Dalal (2000 batch), who at that time was serving as Secretary of the Association and was also a faculty member at the Indian Institute of Technology Guwahati.



Demonstrating exemplary commitment and personal dedication, Dr. Dalal travelled from Guwahati to Jalpaiguri to personally oversee the land registration. In the presence of all land stakeholders and members of the Alumni Association, the registration was completed—quietly, lawfully, and with profound symbolic meaning.

That day was historic.

For the first time since its inception, the Alumni Association became the legal owner of a piece of land located within the campus itself—land procured by alumni, for alumni, and in service of generations of students yet to come.

It was not merely a legal transaction; it was the materialization of belonging, identity, and continuity.

Among students of Jalpaiguri Government Engineering College, the land had long been informally known as “Aranya's location”, a name deeply embedded in campus memory and lore. Whatever its earlier history may have been, the land now assumed a new and enduring identity. It became the official land of the Jalpaiguri Government Engineering College Alumni Association—carrying forward both the legacy of the past and the promise of the future.



This milestone was not the achievement of a single individual, but the outcome of collective resolve and shared purpose. The governing body and active members of the Alumni Association stood united during this defining phase. Among those whose leadership and commitment played a vital role were Gautam Bhaduri (1983), the Founder Secretary of the Association; Dr. Amaresh Dalal, who steered the registration process; and dedicated members such as Bhaskar Dasgupta (1983), Dilip Roy (1984), Nirjhar Chakravorti (2000), Bibhas Bhowmik (1985), Ashok Moulik (1969), and Swarup Bag (2000), whose collective efforts strengthened the Association at a crucial moment in its history.

Above all, this chapter remains inseparable from the extraordinary generosity and vision of Premangshu Ghosh. His selfless contribution made the acquisition of the land possible and left behind a legacy of gratitude, inspiration, and enduring alumni spirit.

The registration of the land in 2013 was not merely an administrative milestone. It was a moment of ownership, identity, and permanence. It affirmed that the Alumni Association had not only roots in shared memories, but also a future firmly anchored within the campus it has always called home.

This was the day the vision found its foundation.



After the completion of the land acquisition, the Alumni Association decided to proceed with the land inauguration. The inauguration was planned for the last weekend of January 2014, marking a significant milestone in the Association's journey.

It was decided that the land would be inaugurated by Premangshu Ghosh, the sponsor of the land. The occasion became a truly historic day, with alumni from different parts of the world gathering on the Jalpaiguri Government Engineering College campus.

The programme was led by the then President of the Alumni Association, Ashok Moulik, along with the Founder Secretary, Gautam Bhaduri, and the then Secretary, Dr. Amaresh Dalal.

From the students' front, the programme was led by Aminul Haque, who was the Assistant General Secretary of the Students' Union at that time and later graduated from Jalpaiguri Government Engineering College in 2014.

The event was coordinated by Kinjal Sengupta (2006) and Nirjhar Chakravorti. The initiative was also strongly supported by several alumni who could not be physically present on campus that day, including Rajib Bhattacharya, Kaushik Mondal (1997) (popularly known as "Omlet" on campus), Bibhas Bhowmik (1985), Dilip Ray (1984) and Saptarshi Sarkar (2013), along with many others who contributed from outside.

The land was prepared for inauguration, and the occasion also featured a seminar attended by dignitaries from across India. Among the prominent speakers was Siddhartha Ghoshal, a 1992 batch alumnus and the then Managing Director of Areva Solar India, who graced the event with his presence.

The land was formally inaugurated by Premangshu Ghosh, who had travelled from Abu Dhabi, UAE, to Jalpaiguri specifically for this occasion. The day marked a defining moment in the history of the Alumni Association.

Following the inauguration, several governing body and core alumni meetings were held—both in person and online—to chart the roadmap for constructing the building. One such key meeting took place in Kolkata during the summer of 2014, where alumni including Gautam Bhaduri, Premangshu Ghosh, Nirjhar Chakravorti, and Kinjal Sengupta gathered at a hotel near Sealdah.

During this meeting, multiple names were proposed for the building, including Rukruka and Kanchenjunga. The name Kanchenjunga, proposed by Premangshu Ghosh, was unanimously accepted.



The decision was deeply symbolic. On clear days, Mount Kanchenjunga is visible from the Jalpaiguri Government Engineering College campus, and the mountain holds profound cultural, geographical, and inspirational significance.

In this broader perspective, the building was aptly named Kanchenjunga—a name that reflects vision, pride, and enduring legacy. Kinjal proposed Alumni Association tagline as 'Driven by Gratitude, Bonded with Nostalgia'.

After the inauguration of the land, a new and far more demanding chapter began for the Jalpaiguri Government Engineering College Alumni Association—the journey from ownership to creation.

The question was no longer whether a building would be built, but how. The Association now faced a series of fundamental challenges: what kind of building should rise on this land, how it should serve students and alumni alike, how funds could be raised without certainty of timelines, and how such a large initiative could be sustained across generations. There was no assured corpus, no fixed roadmap for fundraising, and no precedent to follow. What existed was conviction.



The Alumni Association consciously chose to treat the building initiative as a formal project, applying core project management principles well before such approaches became commonplace in alumni-led initiatives. Strategic discussions included SWOT analysis, phased project planning, cost estimation, risk mitigation, governance structure, and allocation of responsibilities. The uncertainties around fundraising—acknowledged honestly from the outset—were factored into every decision.

The architectural vision for the building, Kanchenjunga, was led by Gautam Bhaduri, whose leadership ensured that the structure would be far more than a symbolic alumni house. Electrical design done by Late Ajay Chatterjee (1983). Under the leadership of Partha Moulik, then President of the Alumni Association, it was decided that the building would include a conference hall, classrooms, guest rooms for visiting alumni and faculty, office space, and a reading room.



The intent was clear: this would not be a private alumni enclave, but a shared institutional asset—a space that could support academic activities, student programs, alumni engagement, and even official college functions when required.

The underlying philosophy was deeply emotional as well as practical. Kanchenjunga was envisioned as a home for alumni, a resource for students, and a neutral, enabling space for the college authorities—a place that belonged to the entire JGEC fraternity. In spirit, it was imagined almost as a temple of collective memory and future aspiration within the campus.

Accordingly, the architectural design took shape as a well-proportioned, sloped-roof, two-storied (G+2) structure, with careful attention given to aesthetics, functionality, and dignity. As a creation of engineers, the building was expected to reflect engineering sensibility, restraint, and purpose—not extravagance.



In parallel, a detailed cost estimation exercise was undertaken, incorporating escalation, contingencies, and long-term inflation assumptions. It was openly acknowledged that fundraising would likely stretch across eight to ten years, progressing in phases rather than through a single surge. With this realism, the total project cost was estimated at approximately ₹2 crore—a substantial amount at the time—arrived at through rigorous planning rather than optimism.

Once the concept and financial framework were ready, the Alumni Association took a bold step. The project was formally announced to the wider JGEC fraternity through presentations, emails, the Association's website, and every available communication channel of that period (2014–2015). For the first time, the Association decided to move beyond campus-centric engagement and reach alumni where they were—across cities and professional hubs.

This marked the beginning of alumni roadshows, a first in the Association's history led by Gautam Bhaduri, Basabendra Nath Ghosh (1991 batch) and Bibhas Bhowmik. Kolkata, Delhi, and Bangalore were identified as cities with strong and consolidated alumni presence. The first roadshow was held in Kolkata in 2015 at the Rotary Club, Tollygunge, where the building plans, philosophy, and long-term vision were presented to an engaged audience.

That gathering proved transformative. Among those present was Snehomoy Roy (1967), a distinguished alumnus from the 1967 batch—the second batch of JGEC—who had served in senior leadership roles across major organizations. His insight reshaped the project's trajectory.



He observed that the proposed alumni building was, in essence, a convention centre—not in size, but in function. It already embodied all the elements of one.

Snehomoy Roy advised that the building be consciously positioned and planned as a convention centre with a self-sustaining financial model, capable of generating its own operational revenue once built. This, he emphasized, was essential for long-term sustainability.

Taking this guidance forward, Nirjhar Chakravorti met Snehomoy Roy at his residence in Kolkata. Under his mentorship, the concept was reworked, refined, and repackaged. The building—later named Kanchenjunga—was reframed through a more structured, corporate-style presentation, articulating its value proposition not only to alumni but to the broader institutional ecosystem.

Armed with this refined narrative, the Alumni Association carried the vision to Bangalore and Delhi. In both cities, alumni responded positively, extending moral support and assurances of financial contribution. Simultaneously, engagement widened through emails, the website, and emerging social platforms such as Facebook and LinkedIn.

Not every response was uncritical. There were questions, doubts, and discreet criticisms. The Association chose to see these not as resistance, but as feedback. Wherever possible, suggestions were absorbed, plans were refined, and communication was strengthened.

Thus, through dialogue, resilience, and collective ownership, the idea of Kanchenjunga gradually took root in the consciousness of the JGEC fraternity. What began as a registered plot of land was now evolving into a shared dream—carefully planned, openly debated, and steadily embraced.

This was how the vision moved from land to life.

Arranging a fund of nearly ₹2 crore was not an easy task. Although assurances were received from alumni through various roadshows and platforms, and there was considerable enthusiasm among the active members, governing body, and the wider Alumni Association, the challenge remained significant. Nevertheless, the journey had to begin somewhere.

Gautam Bhaduri took the initial initiative. He began visiting the campus regularly and formally initiated land measurement and site marking so that the execution phase could commence in a structured manner. With this momentum, a nominal fundraising effort was launched, which enabled the start of soil-cutting activities. A few alumni came forward to contribute at this early stage.



KANCHENJUNGA FROM DREAM TO REALITY

Notably, late Ajay Chatterjee from the 1983 batch and few other alumni initiated contributions, followed by prominent alumnus Alok Basu from the 1968 batch. An NRI based in the Gulf region, Alok Basu contributed a few lakh rupees to kick-start the initiative. Several other alumni also began donating according to their capacity and generosity. It would not be appropriate to name everyone, as many contributed quietly and selflessly.



With these initial funds, the Alumni Association commenced basic groundwork, including soil cutting and preparatory construction activities. Manpower (Labour) management was carried out initially by Nimu da, Alumni Association staff, under the leadership of Alumni Association members such as Bhaskar Dasgupta, Dilip Roy, Bibhas Bhowmik, Rajib Bhattacharya, Nirjhar Chakravorti, Partha Moulik, and Gautam Bhaduri.

The building plan envisioned a main Kanchenjunga building along with a small annex building. The annex was conceived as a prototype—an experimental phase through which the Alumni Association could understand and test project management, coordination, and execution mechanisms. Construction began with this annex structure: a modest G+1 building comprising two rooms.

This small annex building became the first alumni-built structure within the JGEC campus. Although modest in size, it played a crucial role in boosting confidence. It demonstrated that, despite most alumni being spread across India and the world, constructing a building in Jalpaiguri was indeed feasible. The successful completion of this prototype gave the Alumni Association the confidence to move forward with the main project.

Encouraged by this success, construction of the main Kanchenjunga building began during 2017–18. Work on the ground floor commenced, along with gradual fund generation. While progress was steady, the pace remained slow, with much of the support during this phase coming from Alok Basu. Soil cutting continued, followed by the construction of a robust foundation designed for a G+2 structure with a sloped roof. Brickwork for the ground floor also began.

However, toward the last quarter of 2018, fund generation slowed significantly, leading to uncertainty about the project's completion. At this critical juncture, Rajib Bhattacharya (Batch of 2000) stepped forward and took charge of fund mobilisation and construction coordination. His active involvement transformed the trajectory of the project.

Earlier, in 2016, Nirjhar Chakravorti had initiated the creation of a WhatsApp group for alumni coordination. Over time, many alumni were added, and additional groups emerged. These platforms—along with email, Facebook, and LinkedIn—were extensively used to raise awareness about the Kanchenjunga project. Under Rajib Bhattacharya's leadership, fund generation regained momentum.

By the end of 2019, what once seemed impossible became a reality: the Ground floor was completed, and work on the first floor began. Then came the COVID-19 pandemic in March 2020, bringing the world to a standstill. Despite this unprecedented disruption, Rajib Bhattacharya did not halt his efforts. Using digital platforms, he continued to motivate alumni across the globe to contribute and to encourage others to participate.

During this period, Partha Moulik, who had taken over as President of the Alumni Association after Ashok Moulik in 2014, actively mentored and supported Rajib, along with Gautam Bhaduri. Together, they expanded the Association's role beyond construction. Rajib initiated relief efforts for temporary and contractual staff of Jalpaiguri Government Engineering College—particularly canteen and hostel workers and underprivileged persons of the campus—who were severely affected during the lockdown.

Funds were raised to provide monthly essentials, food supplies, winter clothing, and other forms of support. This humanitarian initiative gave the Alumni Association a strong human face and generated widespread goodwill within the JGEC fraternity. As a result, alumni engagement and contributions toward Kanchenjunga increased further.

Even during the lockdown, construction activities continued gradually, providing livelihood support to local construction workers. Through sustained effort and collective commitment, the first floor was fully completed by 2021—a truly historic and emotional milestone in the journey of the Kanchenjunga building.

After the completion of the first floor, the next phase focused on developing essential infrastructure facilities. The first floor was designed as guest accommodation, and Rajib Bhattacharya took the initiative to make these guest rooms fully functional. He encouraged alumni to contribute by sponsoring furniture and essential amenities so that the rooms could be made liveable.



The alumni responded positively and donated generously toward furniture and basic facilities. As a result, the guest rooms gradually became operational, and by the end of 2021, the first-floor guest accommodation was fully functional. Simultaneously, Rajib Bhattacharya also initiated fundraising efforts for the construction of the second floor. Gradually, construction activities began, including structural work and brickwork, and progressed steadily. Dr. Mousam Chatterjee (2004) assumed responsibility as Treasurer and took a leading role in fundraising initiatives, along with Bibhas Bhowmik.

Various alumni members supported this initiative at different phases of the project. While it is not possible to mention all contributors, a few names that come to mind include Kaushik Mandal, Suprovat Chattopadhyay, Nirjhar Ghosh, Romi Roy, Dr. Sandip Jha, Amal Chatterjee, Bitin Basu, Soumyadeep Ghosh, Sanjoy Sarkar, Debashis Chatterjee, Nirmalya Chakravorti, Anuradha Chattaraj, Prabal Majumder, Mrinal Bhattacharyya, Amit Mitra, Barnali Sengupta, Paromita Dutta, Krishna Chandra Sen, Mala De, Sanat Adhikary, Nishit Ranjan Adhikary, Jayanta Dutta Majumder, Mitul Ranjan Chakraborty, Sapath Dutta, Pinaki Ranjan Sanyal, Abhishek Basu, Abhijit Ghoshdastidar, Biplab Kanti Ray, Evelin Bhunia, Anubrata Das, Saurav Majumder, Shyam Sundar Chowdhury and many others. A humble apology is extended if the name of any active contributor or fund raising leader has been inadvertently missed. Though Rajib Bhattacharya was leading the Construction and Project management remotely, there was an active on the ground team at Jalpaiguri who were working relentlessly at campus. The core members of that team were Gautam Bhaduri, Dilip Ray, Bhaskar Dasgupta, Partha Maulik, Nirjhar Ghosh, Partha Halder, Paromita Dutta who were ensuring daily progress and quality of construction.

In August 2022, during the Alumni Association Programme, the Kanchenjunga building was formally inaugurated in collaboration with the college authorities. At that time, the building had reached the G+1 stage, with second-floor construction underway. The inauguration of Kanchenjunga marked a historic moment for the Jalpaiguri Government Engineering College fraternity.

The event was attended by respected professors, retired faculty members, students, alumni from various batches, and college officials. The then Principal of JGEC, Dr. Amitava Ray, was personally present at the ceremony. In the presence of the entire fraternity, the inauguration became a landmark moment in the history of the JGEC Alumni Association and further strengthened the bond between the alumni and the institution. Ex-Principals, Dr. Haripada Bhowmik & Dr. Jatirmoy Jhampati were also present during the inauguration of Kanchenjunga.

Following the inauguration, alumni began actively using the guest rooms. Importantly, visitors coming to the campus for academic and institutional activities—such as campus interviews and official engagements—also started staying at the Kanchenjunga guest facilities, significantly enhancing campus infrastructure and institutional support systems.

Another historic and emotionally significant moment occurred during the COVID-19 lockdown, while first-floor construction was still in progress. During this period, the NAAC accreditation audit—a landmark evaluation for engineering colleges across India—took place at the JGEC campus. The NAAC auditors inspected the Kanchenjunga building and expressed strong appreciation for this alumni-driven initiative.

The auditors not only commended the Kanchenjunga project but also appreciated the broader involvement of alumni in campus development and institutional life. The integration of alumni efforts into JGEC's growth story was specifically highlighted, providing a powerful emotional and motivational boost to the alumni community. This recognition further strengthened alumni confidence and played a significant role in accelerating fund generation for Kanchenjunga and other Alumni Association initiatives.

After the inauguration of Kanchenjunga, while the fund-generation movement continued, construction progressed in parallel under the leadership of Rajib Bhattacharyya, who was serving as the Secretary of the Alumni Association at that time (2021 to 2024). During this period, Alumni Association took proactive initiatives to sustain and accelerate construction by creating a strong on-site resource base within the JGEC campus itself.

JGEC Campus Canteen owner, Papai (Sanjiban Das - son of the popular Bhaida) became one of the key resources during construction activities. Managing construction remotely from outside Jalpaiguri town had proven to be challenging, making this 24x7 localized approach particularly effective.

Under leadership of Rajib guided by Gautam Bhaduri and Partha Moulik and actively supported by esteemed Alumni such as Partha Halder (1993 - current Secretary of Alumni Association), Bhaskar Dasgupta (1983 - current President of Alumni Association), Paramita Dutta (2018), Soumyajit Das (2011), Dr Mousam Chatterjee (2004 batch), a robust network of diverse stakeholders was developed, enabling effective coordination and control of construction activities. This multi-level leadership approach not only ensured the continuity of construction but also strengthened fundraising efforts. Several other alumni also came forward and actively contributed to fundraising, significantly reinforcing the efforts of the Alumni Association.

As a result of these collective initiatives, construction of the second floor was successfully completed during 2023–24. In the meantime, Alumni Association took various initiatives using the Kanchenjunga, such as Scholarship program, students' development seminars, supporting Jyoti's programs, Composting Program (a Sustainability initiative by students led by Nirjhar Chakravorti and guided by Prof Amitava Ray, Principal of JGEC), Readers Corner and support of Campus recruitments etc.

The Alumni Association is now ready to move forward with its next vision: the creation of a state-of-the-art conference hall. Establishing a modern convention facility within the campus has long been a cherished aspiration of the Alumni Association.

In this context, the Alumni Association appeals to the entire JGEC fraternity to continue supporting this major initiative and help transform Kanchenjunga into a hallmark building—one that can be recognized not only across West Bengal but throughout India. Furthermore, the Alumni Association invites all stakeholders to actively utilize the facilities at Kanchenjunga and contribute toward the development of larger infrastructure initiatives, such as an artificial intelligence laboratory, a robotics laboratory, or a sustainability laboratory.

With collective participation, shared ownership, and a long-term vision, Kanchenjunga can emerge as a model for the Indian education system and serve as an inspiring example for institutions around the world.



NB: If we inadvertently miss anyone's name, we are extremely sorry for that. End of the day, it's collective support from ex-students across the globe, along with support from students and respected professors and Principal which made this endeavour successful.

Scholarship Program of the JGEC Alumni Association-a journey from birth to Maturity



Mr Premangshu Ghosh
Electrical, 1979

Mr. Premangshu Ghosh, after graduation, joined the Bhaba Atomic Research Centre (BARC) in Trombay in 1980 as a Trainee Engineer and Scientific Officer. Thereafter, he moved to Reliance Industries (RIL Ltd) Bombay/ Patalganga projects division in 1985 till mid 1992. In 1992 end, he relocated to South Korea in the Offshore engineering division in Hyundai Heavy Industries till 1996. After 5 years in South Korea, he moved to Singapore with Kellogg Brown and Root and worked with other engineering organizations such as ABB Lummas, Foster Wheeler Eastern etc till 2007. After 12 years he moved to Abu Dhabi, UAE, where he joined the ADMA OPCO/ ADNOC OFFSHORE and worked there till 2022. Now though retired, Mr Ghosh is an active Senior member of IEEE and member of CIGRE etc.

The Jalpaiguri Government Engineering College Alumni Association (herein after termed as JGECAA), began the Annual Scholarship for Scholastic Excellence in Year 2013, spearheaded by two ex-students - Mr Bibhas Bhowmik (CE, 1985) and Mr Premangshu Ghosh (EE, 1979); Under the unwavering vision and guidance of Mr Bhowmik and Mr Ghosh respectively.

The JGEC AA, started the academic aid program for students of high calibre or having pecuniary constraints and potential to help achieve their professional goals. What began in year 2013 with two sponsors (Mr Bhowmik and Mr Ghosh), catering two students from each discipline amounting ($\text{₹ } 2 \times 20,000 \times 2 = \text{₹ } 80,000 \text{ PA}$); has now flourished into a full-fledged scholarship program with over 32 sponsors* (both local and globally) to strengthen the student body.

These are in addition to help of the West Bengal Government awarded scholarships scheme which is in existence since long time.

The JGECAA, works tirelessly, to ensure that all ex-alumni from all parts of the world and India come together for the upliftment and expansion of the current study population.

Reference shall be made to Table -1, below, to find the AA members sponsored Scholarship progress trajectory.

It's worth mentioning, that the Scholarship can be initiated/ sponsored by any alumnus/ alumnae, provided these basic requirements are met to the extent practicable

- Selection criteria (merit or need based etc) are open to sponsors choice and can be awarded to any numbers of students from 1st to 4 th year.
Extension of such services may also be considered for Post Graduate aspirants, if any Sponsors so desire.
- Selection is primarily based on direct face to face interview conducted by sole Sponsors or its designated senior members recommended by Sponsor (if not available on site). However, Online selection of candidate by any Sponsor is also encouraged, which is generally commenced during AA AGM scheduled during 1st week of August.
- Stipend amount is open to sponsors choice. No upper limit.
- The fund is collected annually first and then disbursed to successful students directly to respective bank designated accounts.

It is expected that the minimum tenure of scheme shall be continued for long (> 35 years) barring any extreme emergencies.

We at JGEC AA expect much more participation, for such multi-dimensional, benevolent schemes to entire JGEC fraternity within campus and of course outside in near future. Trust, we shall be able to assist more students cross section, with active participation from our ex-students, all over the globe.

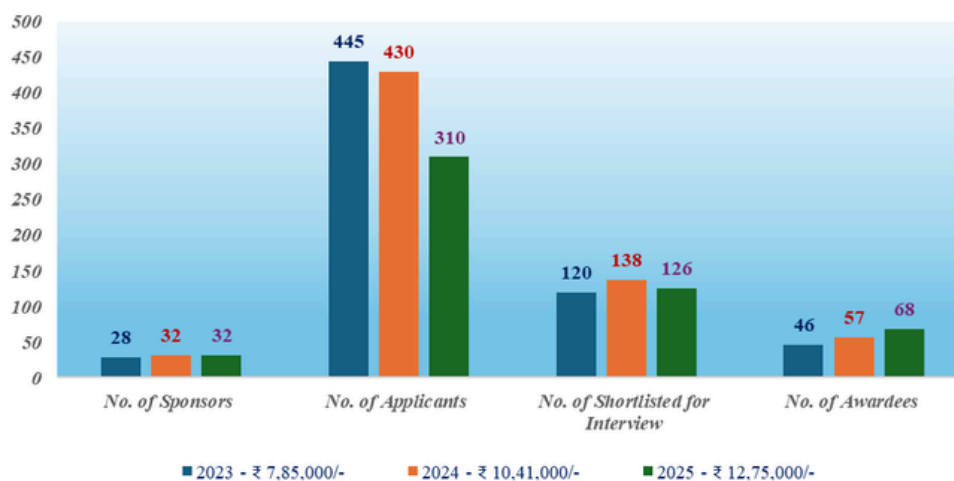
Scholarship Statistics (History)

Year	No. of Sponsors	Scholarship Amount	No. of Awardees	No. of Shortlisted for Interview	No. of Applicants
2025	32	12,95,000/-	70	126	310
2024	32	10,41,000/-	57	138	430
2023	28	7,85,000/-	46	120	445
2022	26	6,71,000/-	41		
2021	22	5,11,000/-	40		
2020	14	3,28,000/-	28		
2019	10	1,97,000/-	15		
2018	8	1,70,000/-	12		
2017	6	1,14,000/-	8		

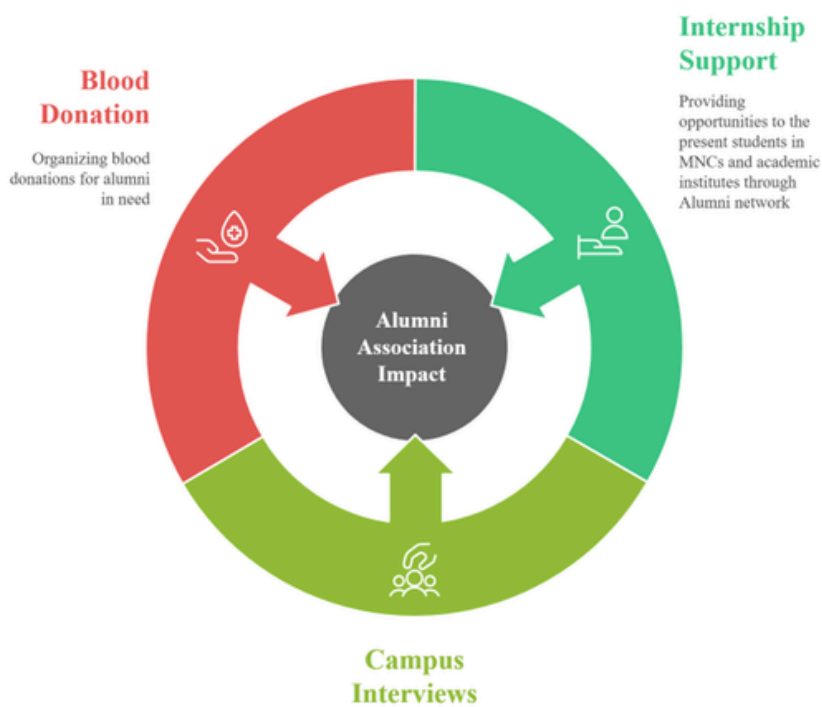
Scholarship Statistics (History)

Year	No. of Sponsors	Scholarship Amount	No. of Awardees	No. of Shortlisted for Interview	No. of Applicants
2016	6	1,36,000/-	10		
2015	5	1,16,000/-	8		
2014	5	1,16,000/-	8		
2013	2	44000/-	4		

SC Scholarship Statistics in last three years



Activities using Alumni Association platform since last year



A NEW TREND : JGEC CAMPUS TO BRIDGE THE GAP



Mr Bibhas Bhowmik
Civil 1985

Sri Bibhas Bhowmik, is a Globetrotter & Technocrat with experiences spanning 35 years in Oil & Gas, Power, Paper, Sea Port , Aluminum Plants and Material handling Facilities & Civil Engineering infrastructure sectors. He has also taught few Industry oriented courses to Elderly working professionals in few Universities of UK at their off-campus unit. Presently he is in expert panel of Mckensy and Company as SME (SUBJECT MATTER EXPERT) and Guest Professor of IIT -Kanpur.

The seminar Day “Bridge the Gap between Campus and Corporate”, held on 21st February 2026, marked a significant milestone in the collaborative journey of Jalpaiguri Government Engineering College and the Jalpaiguri Government Engineering College Alumni Association.



Day long Program is designed to bridge the critical gap between academic preparation and corporate expectations of twenty first century. The seminar aimed to provide students with structured insights into INDUSTRY readiness, Employers demands, and career planning strategies to achieve success in Engineering Career .

The groundwork for this innovative initiative began in September 2025 when Convenor Bibhas Bhowmik originated a detailed Standard Operating Procedure (SOP). Based on this original framework and value-added input from Seminar Vertical alumni team, a comprehensive Power Point presentation was meticulously prepared by few students support team members Akash Das, Rupsa Dey, and Kuntal Roy, with assistance from Pradip Kumar Roy, incorporating visuals of the iconic Kanchenjunga Building and the college campus. The presentation underwent several iterations to ensure clarity and institutional alignment before being forwarded to the Secretary and President of The JGECAA for final submission and official record. Parallely, a Google Form was circulated among students to collect feedback on preferred seminar topics, ensuring that the event was responsive to students’ aspirations. The approval and coordination process extended over two months, with the final date selected strategically in consideration of academic schedules and GATE 2026.



The seminar featured distinguished speakers: Dr. Jyotirmoy Banerjee (Chief Human Resources Officer, Hindustan Construction Co. Ltd.), Evelyn Bhunia (Bechtel India Pvt. Ltd., core domain Specialist), and Ms. Indrani Ghosh (Cognizant Technology Solutions, Senior Employee) who shared invaluable industry perspectives. Sponsorship support was led by Mackintosh Burn Limited as Platinum Sponsor, with other event sponsors ADDCool Trades and Services, Jayanta Dutta Majumdar (Indus Wealth), and PSP Fintech Consultants.

Professor Ayan Pramanik played a pivotal role via liaising with the Heads of all six Departments to finalize Seminar Date: college logistics and venue arrangements. The Language Lab was designated as the primary venue, while live projection was arranged at the Kanchenjunga Building ground floor hall to maximize participation and realising utilisation of this Convention Centre.



We are grateful to few alumni, who acted as great patrons of the event . Webcast technical arrangements were efficiently managed by Arijit, Akash, Rupsa, and Rahul. Different promotional creatives were developed by Akash and Rupsa; certificates and tokens of appreciation were designed with digital authentication from Principal Amitava Roy and President of The JGECAA ; and media outreach was coordinated by Convenor Bibhas Bhowmik and Rupsa Dey. M/s Good Ricke assisted The JGECAA for providing extensive media support .

The execution day reflected months of disciplined nitty-gritty planning and extensive teamwork. The sessions were witnessed by enthusiastic student participation, active engagement during interactions, and meaningful dialogue between alumni and undergraduates. The hybrid format—hosted physically and webcast via Google Meet—ensured inclusive outreach first time in CAMPUS



Down Memory Lane- My memories with JGEC

It was an idle morning in November 1961, when I set out to join Jalpaiguri Engineering College (JEC). The journey was a tedious one, almost 24 hours journey from Kolkata, involving 45 minutes of ferry service, as there was no Farakka Bridge in those pre-historic days. Despite the trouble, I was in high spirits as it was my first job and a lucrative one for me.

As for the location of JEC, it was near Danguajhar Tea Garden; so, I headed for that area, but to my dismay, found no trace of a college; only a Polytechnic Institute existed there. Bewildered, I looked around and was struck by the snow-white heavenly view of Kanchanjanga in the backdrop of a clear blue sky; it was really enchanting. But then I must not forget the work at hand: I have to look for an engineering college first; so, I kept strolling near the polytechnic, when a board with “Jalpaiguri Engg. College” written on it, came to my notice.



Prof Subhas Chandra Bose (Retd)
Deptt of Mathematics
Jalpaiguri Govt Engg College (1961-1984)

Below, was a 2- room block rented out to JEC for provisional use as an office. I entered there and was greeted there by two persons: 1. Saday babu, a short bulky aged man and 2. Deb Kumar Mukherjee, a smart handsome young man. They were the Office Superintendent and Asst Superintendent of the upcoming Engg. College. They held out to me, a white paper to write a short joining report on it, which I did. Thus I, as a lecturer in Mathematics, became the first full time faculty member to join JEC. It was 13 November 1961. The same day, after a while, a bald-headed Asokananda Chatterjee appeared from no- where to join as a lecturer in English.

Then it was a problem for us, where to put up. The principal of the college, hon'ble Prof N C Bose, an upright tall old man, whose features commanded respect took this problem in right earnest. The same day, he and Saday Babu accompanied us (myself and Chatterjee) to Jalpaiguri town in search of a mess and successfully lodged us in a queer Achal Mess. There we stayed about a month or so and thereafter shifted to another queer- named Goshala Mess, near the college area.

The first batch of students, about forty-five enrolled, few of them local but mostly from outside; they also needed a hostel which was non- existing then. They were also boarded in the same Goshala Mess, which was then renamed JECSAR (Jalpaiguri Engg. College Students Approved Residence).

Coming days saw various challenges: Land acquirement, construction of college building, hostels, workshops and housing complex for teachers and staff members; everything was absent then. As such; we, teachers and students alike had to bear the brunt of it. There was no Maskalaibari bridge, so the students had to go round half of the globe to reach A.C. College laboratory to do practical. So also the teachers had to bear the same hardships, going to A.C. College. There were no engineering teachers then; Thank God, the 1st year curriculum includes only science, mathematics and humanities; and no engineering subjects. But there was a sense of urgency regarding construction of college building, hostel, workshops etc. and recruitment of engineering teachers; because next batch of students would be coming soon. In that sense of urgency, we saw the coming up of low-cost housing made up of hollow bricks in 1962, an idea mooted by Principal A.C. Roy of Bengal Engg. College, Shibpur, Calcutta. These low-cost housings were earmarked for residential purposes. At the same time, construction of a college building and a hostel were taken up in a war-time haste. By 1964, a portion of the college building was ready, while the construction of hostel 1 was still in progress. By this time two more batches of students have poured in and recruitments of teachers in all the departments were in full swing. As the hostel was not ready, the students were boarded provisionally in the portion of the college building that was ready.



Those were the happy days for all of us, teachers and students alike. We were happy to have so many friends and colleagues; arranging frequent get-togethers and annual picnics; and the students happy with so many celebrations all the year round, apart from their studies; organizing sports and annual socials. Annual Social, organized by the students in those days, lasted 2/3 days with extreme joy and enthusiasm; bringing in luminaries like Ali Akbar Khan, V.G. Jog, Hemanta Mukhopadhyai, Debabrata Biswas and so on so forth.

Then came the year 1965, which was marked by a tragedy that shook us to the marrow. If I correctly remember, it was 10th November when two bright young boys, two of our loving students Sujit Ghosh and Kajal Banerjee (if my memory is not failing) drowned in Karala River while swimming.

60 years down; all the water that has flown through Karala since then could not wash away the memory of this tragedy; a tragedy that shook JEC in its formative years.

No matter how great is a tragedy, 'Time' is a great healer. In a year or so, the memory gradually receded to the background and came the days of business.

Now that the college is running with full-fledged 5- year batches; with teachers, though not adequate in number, can somehow pull on the teachings in all the departments, it is time to think of an administrative body, a governing body (G.B.). Such a G.B. was formed with representatives from all relevant constituencies and a representative from the govt. of W.B. This included, among others, two from teachers' constituency, two from students' constituency, two renowned tea magnates of North Bengal (S.P. Roy and B.C. Ghosh), two engg. College Principals (Prof. A.C. Roy of B.E. College and Prof. N.C. Bose of JEC). And the govt. representative in the G.B. was Sri Khagendra Nath Dasgupta, a minister from Congress Party.

One of our teacher representatives was an outspoken Asok Mukherjee (?) of mech. engg. department. In a G.B. meeting (probably the 1st), he made so virulent an attack on the previous administration, that a red-faced Principal N.C. Bose (of JEC) sent a job-termination notice to him, the very next day, and he was sacked; not only he, another teacher (S. N. Bhattacharya of the deptt. Of Humanities) who was believed to be the kingpin in the selection of Mukherjee from the Teacher Constituency, was also sacked. However, his termination letter was later recalled when the authorities felt he was wrongfully implicated. As for Mukherjee, he was not the least bothered, as he came on lien from Jadavpur Engg. College and went back to his parent institution.

Nevertheless, this unsavoury turn of events had a chilling effect on the teacher constituency and the community as a whole. As a fall- out of this, the engineering departments of JEC, already in dire need for more teachers, remained perennially understaffed. This caused discontent among the JEC- students; and they felt: the malaise afflicting the college will not be over unless the Govt. takes over the administration. So, they started agitating over this and some other allied demands. 2/3 years of unrest passed by; and then the victory came their way. The college (JEC) becomes a govt. college, renamed Jalpaiguri Govt. Engg. College (JGEC) on 15 February, 1968. The students heaved a sigh of relief.

Samanta Ghosh (?) one of the students spear- heading their movement announces: Now that the college is taken over by the govt. and our primary demand is fulfilled, we are happy and want to settle down to our studies in peace. But peace eludes more often than not. Fortune had other designs. Politics came in the way. Charu Majumdar's 'Nakshal Movement' swayed the student community in West Bengal, affecting the academic atmosphere everywhere, particularly in JGEC where it was in its cradle.

Anything that starts, whether it meets success or failure, comes to an end. Nakshal Movement, after causing considerable damage to life and property, fizzled out in due course, but left a socio-economic imprint on the young mind, which later changed the political scenario in the state of West Bengal.

By this time, adolescent period is over for JGEC; quite a few batches of students have come out and they are (most of them) are highly placed, making the college proud. About the same time, the teachers of the college were sent on deputation for higher learnings to different IIT's under the UGS sponsored Quality Improvement Programme (QIP). In 3 yrs. time, they come back better- equipped for imparting education; thus the college grew in stature.



As for myself, I went to Delhi IIT in 1975, where I did my Ph.D in 'Functional Analysis', an abstract area of mathematics and came back to JGEC in 1978. But Functional Analysis is not an area taught in engg colleges like JGEC; it is better related to university education. So, getting an offer from Manipur University, I joined there as Associate Prof of mathematics in 1984 and then back to West Bengal in 1987; joined North Bengal University (NBU) as Professor of mathematics where I stayed till my retirement in 2003.

I am now 87+. While I am completing this write-up to-day (13 Nov. 2025). I recall with nostalgic feeling in my heart, that this is the same day in 1961, when I joined JEC as a young teacher.

With love and blessings to all the alumni of JGEC, I remain.

In the Pursuit of History- First era of our college

-By Batch 1971

***"We never realize the value of something in our life until it becomes a memory.
A moment lasts for seconds but the memories last forever"***

August 7 of 1961 is a memorable day for all of us who started new journey of life in the search of educational goal after schooling. A new college named Jalpaiguri Engineering College was born on that day. ... 5th oldest engineering college of our beloved state, West Bengal.

Our favourite college, in the year of 2021, stepped into coveted Diamond Jubilee year as coal of Bengal is turned into diamond under pressure and time.

First ever private engineering college of West Bengal started her maiden journey with approximately 60 number of students being distributed in 3 departments. Today, on diamond jubilee year, our favourite college stood held her head high for her rich mantle of technical excellence and being migrated from private status, affiliated to University of North Bengal, to the institute of complete autonomy. It traversed a path of glorious technological advancement changing the world and taught students how to live in it and it is now a prestigious NAAC Accredited institution comprising of 6 undergraduate and 2 postgraduate departments. Gem of students are born from the golden womb and manifested on her lap to spread the knowledge and wisdom in every corner of the Universe.

There lies history behind every creation. History does not mean that it should be century old, a 60-year-old reality is also a history. Everyone who are related to a particular creation should know the history. History, if you don't know, then you know nothing. A leaf that you are, and you don't even know that you are a part of the tree. So, everyone who studied or taught; organised or maintained workshop and laboratories with their best effort and kept the thing moving, they should know the history of that era and the history of political and administrative involvement in the betterment of the institution in post era as well.

India earned freedom at the cost of partition of Bengal and Punjab. The decision of British drawing a line was not an administrative formality, but it was a matter of life and death of our countrymen. Millions of people were displaced, fled and migrated to various Parts of Bengal in Kolkata and her suburbs; scattered in various districts in North Bengal. Bengal had, during partition, 2 nos of engineering college to her credit named Bengal Engineering College, Shibpur (presently known as Indian Institute of Engineering, Science and Technology) and Faculty of Engineering, Jadavpur University, in addition to India's first premier IIT, Kharagpur which was established in 1951.

India cannot remain insulated from development and understands that future growth depends on greater technical skill of development. Thus West Bengal was awarded with an engineering college by the centre at Durgapur under Burdwan University (Presently known as National Institute of Technology, Durgapur).

Our college was established, exactly after one year, being initiated by 2 great souls of Jalpaiguri, respected late Birendra Chandra Ghosh and honourable late Satyendra Prasad Roy. These two gentlemen, with the interest of development, made utmost effort together with other likeminded people and established the private engineering college in close proximity to Danguajhar Tea Estate, situated north of Jalpaiguri Polytechnique. Their minds got filled with ecstasies as they realised their dream came true and acknowledged the generous participation of honourable Mr Khagendra Nath Dasgupta, the then Minister-in-charge of PWD, for the assistance of development of the college.

Our college started initially with 3 departments having poultry students of 60 in 1961 with scanty arrangement of accommodation of students. Students of first batch suffered the teething problem of administration and stayed scattered in the house of relatives in town area, or some groups were housed in the hostel of poly-technique situated adjacent to our premises.



West Bengal Board of Secondary Education introduced new curriculum of Higher Secondary Education in 1960. It discarded old Matriculation system where students were allowed to take admission in engineering colleges after qualifying 2 years Intermediate of Science Course, after matriculation. But under new system, students passed with science stream were eligible for admission into 5 Yrs. Engineering degree course. Many schools could not adopt the new system due to some unavoidable reason. Students who passed out from those schools after matriculation; had to go for one-year Pre-university course in any degree colleges so as to qualify for Joint Entrance test. Board of joint entrance for engineering was formed in 1962 and eligibility criteria of admission to engineering college was altered and students were being admitted through Joint Engineering Entrance test only. Our college was initially affiliated to Calcutta University and affiliation was changed to University of North Bengal in 1963 after this new university was established in 1962.

The requisite infrastructures like Main building, hostels and workshops etc were under construction in initial time so students were being sent to Jadavpur and Shibpur for necessary educational teaching assistance. It is known, by 1962, a few number of workshops were made ready to accommodate some students for boarding and balance students were housed in nearby Patkata colony on rent. Fresh batch of students, admitted in 1963, were accommodated temporarily in houses situated close to Danguajhar Post Office. Hostel No 1, presently known as P.C. Roy Hall, meant for 2nd year students, was completed in 1963 where students of batches of 1961, 62 & 63 were accommodated that provided with a sigh of reliefs. Hostel no 2, presently known as J.C. Bose Hall, meant for 1st year students, was made partially ready in 1965. All students who were staying in workshop and Danguajhar were offered accommodation in newly constructed hostel no 2. As some part of hostel No 2 was incomplete, Mr. Naresh Chandra Bose, the then Principal, was kind enough to accommodate freshers of 1965 batch in some part of main college building. Two no of hostels were constructed with 75 no each of 3 seated rooms but due to scarcity of adequate accommodation, all rooms were converted into 4-seater, causing huge inconvenience to students. College administration decided to construct some low-cost houses recognised as Approved Residence (AR) to have makeshift arrangement for incoming students of 1966 batch and also for some workshop employees.

Students staying in AR were extended food facility from the mess being run in hostel no 1. Gradually, in this fashion, 600 odd number of students started their memorable hostel life within the premises of college. We know that light displaces darkness but we used to experience, after 11 pm every night, that darkness would cover the entire campus because our college did not enjoy the facility of power supply from State Electricity Board but had in-house Generating power system being run by DG set which has taught us to light a candle in the darkest moment of our life....a great lesson we learnt in our early life to cope up with odds.

First batch of 1961, amidst of constraints and persistent difficulties, graduated in 1966 with flying colour, despite huge constraints and difficulties faced during their formative year of engineering career and stepped out of college campus for ever in the pursuit of new professional life.

The history of technical development and of technical education are naturally inseparable. The industrial development of any state has to create a demand of person with necessary technical education. Let us rewind to see the political and economic scenario of state as well as our country. Sino-Indian full-fledged war broke out in 1962, immediately after formation of our college, and subsequent Indo-Pak war was held in 1965, followed by Food movement in 1966 due to food crisis and sky-rocketing food prices that worsened the health of industry as well as that of state and put the development curve of industry downward.

Any newly established college cannot escape the lashes of precarious health condition of state. Our college, in fact, faced huge financial crunch due to dearth of fund and with no governmental aid. The entire support system was on the verge of collapse as there was no other means to generate fund. Capitation fee was never an instrument of fund collection during those time. Transforming our college of higher learning into a centre of excellence became a remote problem.

Lack of teaching staff, poor quality of laboratory and workshop. dearth of good books in library, poorer quality of canteen facilities and insufficient infrastructures became the recipe of dejection and worries amongst final year students.

The stresses that come with academia are enough to think about future and opt for agitation for the betterment of the institution they love.

So, agitation could not be averted, agitation for development of our college saw the light in mid of year 1967. This spark ignited the same feelings amongst the students of junior years as they recognised the common goal that led to a consensus agitation. Students formed an action committee so that proper planning can be done in order to lead the agitation in right direction.

It was not a simple but a complex agitation due to status of private educational institution and prevailing examination system of the college. Our college had orthodox annual examination system of promotion. Student would be declared as Fail if anyone failed to secure pass mark in more than two subjects and student would be declared as CNR (Can Not Repeat) if one fails to qualify twice in one class and would be expelled from college. All these criteria put students in dilemma as well as they became wary about future. Everything becomes good if it is started with good and noble intention. Initially, all faculties were in dilemma and had doubt about agitation but in the passage of time as agitation gained momentum, they got convinced and extended moral support... was the sunny sign of agitation.

In due course of time and in right opportune moment, a Charter of Demand was placed on the desk of Principal with a request to take up the matter with higher competent authority for kind consideration.

Thing does not happen in the way one desires. The competent authority did not pay much heed to our charter of demand and their unwillingness hinted about their incompetency to tackle the issue. Moreover, they could not manage to get any sanction from any authority.

Time waits for none and moves in its own course. 2nd batch graduated out and new batch of students was admitted in 1967. Political scenario of West Bengal also metamorphosed and hailed a new form of Government. United Front took over the rein in March 1967.

It is a fact that progress does not come without struggle, but this struggle should be moral. Our agitation had merit and moral support, so it gathered momentum. Agenda of agitation was changed and a new demand cropped up for taking over the college by the government which proved to be a strategic one and this move earned support of teaching and nonteaching staff of the college. Our rational minded Principal advised to take up the matter with VC of the university. Student delegation met respected VC, Mr A.C Roy, a splendid educationist, who assessed the gravity of the situation and agreed to provide support in raising the demand in the education department of newly formed government but demand for acquisition of institution by government should be raised through students.

Under the pretext, first non-political Student Union of the college was formed to represent our issue to the competent authority. President and Vice President of the union were chosen from Final year. General Secretary and Asst Secretary were elected from 4th year, and other representatives were chosen from the students of remaining years. An agitation, may be small or big with noble cause, needs support from people of each layer of society for its success. So, our Union felt the necessity to seek support from various layer of people from Jalpaiguri town, in addition to supports already acquired from our own staff. Accordingly, our agitation was placed tactfully in wider galaxy beyond our college boundary. Various walls of college and Jalpaiguri town were found to be pasted with posters and leaflets were distributed amongst general citizens as well as to the students of other colleges to make them aware of our legitimate demand. Many rallies were being made through different streets of Town with an aim to meet with leaders or MLAs to get our desired supports.

A group was formed to keep track on the possible movement of any big leader of state in nearby area so that we could place our demand before him. It appeared to be very simple, today, in digital age but it was an uphill task in those periods. The greatest force that is derived from mind ... a Will force creates havoc in life. Mr Nani Bhattacharyya, MLA from Alipurduar was traced by our group in Jalpaiguri Road Station

and he was kind enough to visit our college and listened to our demand. Our agitation continued and Mr Hare Krishna Konar, the then minister of West Bengal government who came to attend a political meeting in Jalpaiguri and representatives of our union convinced him with great effort to visit our college and to take a cognizance of our demand.

It was evident and crystal clear to us that nothing could be materialised if sufficient pressure is not thrust on ruling government. Action committee and union took a unanimous decision to have one day token hunger strike before opting for indefinite one if situation demands. Two of efficient representatives were sent to Writers' Buildings, Kolkata to brief them about our demand and try to open an avenue for effective discussion on our issue in higher level. Our college authority, Principal of college, Vice Chancellor of the university, District Commissioner and Superintendent of Police etc were served notice for their information. Local reporters and political leaders were also taken into the confidence by briefing our future action plan. Our agitation came out of its incubation and became full-fledged one.

End of September 1967, a call for Dawn-To-Dusk mass hunger strike for one day was performed- All canteens, big or small, remained closed. All teaching and non-teaching staff including family of our Day-scholar students kept their kitchens shut to show solidarity for our agitation and our college campus became the nucleus of greater agitation.

On the same day, our Principal put his effort to bring college authority and students on the same table for amicable discussion on the issue, but the authority during discussion, failed to show any rays of hope to us, i.e. Students representatives present in discussion declared categorically to go for indefinite hunger strike till death. This was held on the ground floor of college building. Selected group of students were entrusted with the duty for looking after them in turn day and night. College authority was requested to intimate to the office of VC and other administration. Infinite hunger strike of a group of 18 students until death started. Students, participated in hunger strike, were chosen from 1963 to 1966 batches, leaving apart the students from 1967 batch as they were freshers in the college. Professors and other staff of our college used to be around during hunger strike to raise the morale of students participating in agitation.

On the other hand, in the Writers' Buildings, the then official secretariat of the Government, our representatives successfully met Joint Secretary (Technical Education), Secretary Education and also met Mr Jyoti Bhattacharya, the then Education Minister. They were able to brief efficiently about our demand and future path of our agitation.

Hunger strike, our non-violence resistance, continued more than 48 hours. All students were under constant medical check-up duly arranged by college authority. One student from 4th year fell sick. Dr Anupam Sen, Principal and other teaching staff advised him to withdraw due to his bad health condition and remaining 17 students continued.

Beauty lies in the mind when it surfaces you never know. It is what we realised in the eventide of 3rd day. Our non-violence resistance was able to create cohesiveness and oneness amongst entire community starting from family members of teaching staff who came to place of hunger strike to share feelings, sentiments and unity in order to absorb the suffering and agony of the students.

“Little waves arise in many directions: but one swallows up all the others and sweeps over society. That wave does this which has more character at its back”. - Swamiji

After passing of 3 critical days i.e valuable 72hrs of life , a faint light of hope arrived that government was set to sit for discussion with our representatives. The subject meeting was presided over by Mr Jyoti Basu, Dy Chief Minister, Mr Hare Krishna Konar, Mr Amar Prasad Chakraborty and some secretaries. Long marathon discussion was held, and the matter was discussed at length with all pros and cons related to development of some possible administrative problem that might affect governance of state administration. Government authority declared that subject demand cannot be sorted out now due to some administrative difficulties but assured us that our demand has been encoded for future consideration and advised us to withdraw our agitation with immediate effect.

Our 84 hours long hunger strike came to an end after positive assurance given by the competent authority. Indefinite hunger strike was withdrawn but not the spirit of agitation. Students were alert and observant about political activity, in and around, that might help us. Autumn arrived with her natural fragrance and announced the emergence of goddess Durga. We came to know, immediately after puja vacation, that Mr Jyoti Basu, Deputy Chief Minister would visit Jalpaiguri in connection with some administrative issue. Student union decided to meet Mr Basu. We conducted a peaceful procession and flocked together in front of Circuit House to meet the Deputy Chief Minister. Deputy Commissioner and Deputy Superintendent of Police requested Deputy Chief Minister to meet us, but he declined to discuss anything with us. But we were firme and adamant. We refused to leave the spot. Perhaps, Almighty had different plan, heavy downpour started. We all students stood silently under incessant rain like statue. May be, our valley of silence like attitude softened and touched Dy CM who came out suddenly and rebuked us for getting drenched in heavy rain, lest we should catch cold. He then came to the point of our demand and narrated categorically the inability of present government to tackle our demand due to some administrative and financial constraints. But Government is well aware of the demand and assured to take up in future. We became perturbed and left the place thinking as it was neither victory nor defeat but only assurance. Life is a silent teacher. We kept faith...

Faith is the bird that feels the light and sings when the dawn is still dark ... Tagore

At last fortune smiled on us. All of a sudden, on 20th Nov, 1967, All India Radio broadcast in the evening local news that Government of West Bengal decided to take our college under the arm of First United Front Government. Official confirmation of the news reached to college authority next day through Telegram.

Jalpaiguri Engineering College (JEC) Renamed as Jalpaiguri Govt Engineering College (JGEC)

Environment of entire college premises was electrified and was inundated with joy and happiness. It was a monumental victory. The victory was about our effort laid on the foundation of conviction & determination and when it paid off, everyone came to know the history. Pure joy & ecstasy though it looks like an ephemeral emotion. We can very well hold on it, and we can relish in future to come as we do today, even after 5 decades.

Ironically, our agitation didn't end there. How can it be... agitation is opposite to stagnation, and it does not have any pause. United front government put forward their resignation on the desk of Governor on 21.11.67, immediately next day after acquisition of our college. Governor dismissed UF cabinet and let other group to form next cabinet and government formed by them was crumbled within 3 months and state came under Governor's rule since then, for next one year. Political instability and intolerance amongst political leaders scripted different message on the call about future of the state. Democracy and discontent gradually became the growing crisis of governability. Nothing could be achieved for the benefit of our college, during these 15 months, barring only disbursing of salaries to staff of our college. Students of 1963 batch graduated and stepped in profession world to fulfil their dream.

Time is a typical mover who does not allow anything to be at rest. And it didn't allow us a calm course of existence. Jalpaiguri suffered severe and unprecedented fury of nature. Jalpaiguri district is usual prey to flood hazard. The extensive damages caused due to river Tista were unprecedented in magnitude. Embankments, saviour of Jalpaiguri town, were breached in many locations causing huge inundation in town and adjacent areas. Our college campus was no exception and became victims of flood and college authority compensated for the loss for the affected persons staying in ground floor of hostels.

After 15 months, on 25th Feb 1968, second United Front Cabinet assumed office led by again Mr Ajay Mukherjee. During the period, significant political violence was noticed. Clashes between parties, coalition in fighting and conflict made the political scenario of West Bengal very gloomy. Repercussion of this misgovernance also did not spare our college. Our college was functioning without Principal. College Administration was being run by officiating faculties. Under this situation, students of 1964 batch graduated out in search of their dreams.

Bengal in late sixties was burning due to side effect of peasant movement being originated from Naxalbari in 1967. Young generation was blooming in confusion and frustration. Situation was chaotic and volatile. The normal functioning of various colleges and universities were affected, and regular classes were suspended that led to stalemate condition of many colleges in West Bengal. Under such chaotic and turmoil condition, 2nd United Front also collapsed, and President's rule came again in force.

The decade 1970 was a strange decade for villages, towns, city and for state, marked by intense violence and gruesome killings around. People were horrified and terrorised as if youth of West Bengal have uncanny romantic involvement with violence. Education system was attacked in the plea of 'bourgeois system'. Colleges and universities became epicentres of this movement resulting colleges to close for Sine die. Our college also became victim of this movement. Newly appointed Principal of our college left without notice who could not bear the heat of the movement. Our college became orphan overnight, and college was closed but hostel remained open. Situation worsened. General students started living with fear and anxieties. Distrust and disbelief prevailed around. Environment was conducive for everything but education. Despair and frustration were our companion. In such depressing environment, our leaders kept focus to north and representatives of our union met Advisor to Governor with a request of reopening the college.

Mr R Jakuma, the then administrator, was appointed as Director for running the college smoothly. He drew his own plan of action with priorities. He decided to isolate students first, so he closed hostel as per priority of his action plan and next, he sent all students back to their native places. After making necessary arrangement, he started calling back, students in alternate batch, to college and conducted examination, though session was delayed about nearly 8 months. Slowly but gradually normalcy was restored back to campus as a result of able and efficient governance of Mr R Jakuma. The struggle of that era didn't end there. Those who played major role and were emotionally attached kept their focus on the progress of the college through Alumni.

First era that was full of visionaries who had strong conviction about learning and commitment for development. Today, even after 5 decades, when they find their college is in right track of progress, their happiness knows no bound to find their dream came true. Their happiness that lies on the satisfaction of constructive effort they pursued.... They didn't let anything happen but make it happen.

Over the time, fortune of college began to change gradually, and new dignities began to fall like flowers on her lap. Our college began to be adorned with new glory. Our college achieved growth not only in volume but also adding more substances to the core. In 2012, UGC awarded our college autonomous status under government of West Bengal. After getting recognition from NBA (National Board of Accreditation), our college bagged the valuable recognition on 23.02.21. Our college added another feather on her cap on 30.09.21 .. college was decorated with prestigious CENTER OF EXCELLENCE(COE) which will also do activity related to Cyber protection, marked as only institution in North Bengal to be awarded this honour.

It is a pleasure to know that the students of our college are successfully grabbing the employment opportunity in the company they desired. and about 70% of students got employment. Our kudos to placement cell. We do expect that the authority will do more to promote our college and pave the way for further improvement so that it becomes one of the leading colleges in India. All students of first era are today very proud to see that the fruits of their struggle are leading their thoughts towards perfection

***The history of this world is the history of the few people who had faith in themselves... You can do anything. You fail only when you do not strive sufficiently to manifest infinite power-
Swamiji***

The identity of our stalwarts whom we adore from bottom of our heart are kept undisclosed deliberately as it was a united movement. Everyone participated in same vigour and veracity in this history of manifestation. We don't prefer to measure participation of different member in different scale as everybody strived for success and fulfilment.... a complete movement of unison and a successful harvest from a joint struggle for meaningful existence. Many of them are not with us but they are remembered by us with love and respect.

A Homage from the students of the first era on the landmark of "Diamond Jubilee Year in the odyssey of their beloved college.

(This history was presented by the Batch 1971 and was written in 2021)

Dialogues That Matter

Face to face with



Debashis Chatterjee (Mechanical, 1995)

Vice President & Chief Operating Officer
ADDCool Trade & Services

Debashish Chatterjee has worked in many reputed organizations like Philips, Stirling Cryogenics BV, ELGi Equipments Limited etc in different part of India and abroad. After working for more than 17 years he started his own entrepreneurship in the name of ADDCool, which operates in the Cryogenic Segment.

From Circuits to Startups

1. At what point in your engineering journey did you realize that solving technical problems alone would not cut it, and that the real puzzle was solving market problems? Was there a particular moment, client, crisis, or “cup of coffee” that triggered this shift?

Reply: I was in Bangladesh during 2014, heading the business of M/S ELGi Equipments Ltd., for Southeast Asia. Every evening, I used to get a call from some Customer in India, who were not getting sufficient After-Sales Support from my previous company (in Cryogenics). This initially pissed me off, as I had resigned long back from that Organization. However, then it struck suddenly that this is huge potential market which I was missing. I have been dreaming of my own company since a long time. Then I realized, it was time. We started as a Service Provider to several Cryogenics Users and then gradually scaled up to what we are today.

2. What was the toughest technical habit you had to unlearn on your way to becoming an entrepreneur? And on the flip side, what was the hardest new skill or mindset you had to acquire?

Reply: I did not. No knowledge / habit is so cheap to unlearn. In fact, I added on. The most difficult skill that I had to acquire was Man-Management. When you do it for others, you hardly can feel the stakes. Here, perfect balance was necessary, creating a healthy team along with a competitive attitude.

Engineering the Human Factor

3. As someone with a systemic mindset, how did you go about designing the “human operating system” of your company? Did you borrow from any standard playbook, or was it more trial, error, and human debugging?

Reply: My Philosophy in life has been to grow with my surroundings. My surroundings include my employees, my vendors, my customers and even my neighbors. No growth is meaningful, unless the surroundings grow with you. This triggered my Team to understand that they are supported by a senior Team Member, who credits them for successes and accepts the responsibility for every failure.

4. In the early days, you were probably chief problem-solver, fire-fighter, and everything else rolled into one. As the company grew, how did you manage the transition from being the “lead solver” to becoming the “lead delegator”?

Reply: **Delegation can only happen when you are yourself capable to solve a problem / situation or can motivate others to do it. Unless you know the art of yourself to resolve a problem / situation (even if you are not doing it yourself), you cannot really delegate. In such cases, where you are also a part of the problem solving, delegation comes through self-motivation and my Team is motivated. Thus, we could survive in this competitive market for 12 years now.**

Wisdom for the Brave

5. For an engineer who has spent 15–20 years in the corporate world, what is the most underrated advantage they carry into entrepreneurship, perhaps without even realizing it?

Reply: **For an engineer with 15-20 years in the corporate world, the most underrated advantage is: 'The zeal to set a Goal that is specific, measurable, achievable, realistic and timely, and achieve it'. Every engineer in the corporate world is serving to provide meaningful outputs in a measurable and timely manner. If any one of the adjectives out of the five are not followed, the effort may become ineffective. This is exactly the advantage we carry in the backpack.**

6. What is one industry-wide “accepted truth” that you believe is outdated, overrated, or just plain wrong?

Reply: **That statement "Emotion has no place in the industry". I firmly believe that passion comes from emotions. A person without emotions (or less emotions) will not carry the passion to achieve. In my organization, we allow flamboyance of emotions. Thus, there is no dearth of new ideas.**

Advice to the Next Batch

7. If you were addressing a room full of fresh graduates from our college, what is the one lesson you would most want to leave them with before they step out into the real world?

Reply: **Every work is Entrepreneurship. So, all of us are into business. You reap as you sow. Courage is the most valuable thing in this world, which we all stagger upon, at some point in time. Work on it.**

The Important Questions, Obviously

8. Finally, we all know and admire you as a truly multi-talented personality — a fantastic singer, a valued alumni contributor, and someone inexplicably popular with a certain species we shall refrain from naming here for diplomatic reasons. Naturally, the question is: how on earth do you even manage your time?

Reply: **Thank you for the compliments. I still have miles to go. Come to me at the end of the journey and I shall let you know.**

Installation of AC in Kanchanjanga- an initiative of 1991 batch

Mr Basabendra Nath Ghosh
Mechanical -1991



It was not a dream, a midsummer night indeed. Fellow members of 1991 batch were leisurely chatting in their own group platform during early 2025. Topic was Kanchenjunga, not the majestic Himalayan peak. It is their second home at campus, slowly and gradually taking a real shape, that was once a dream a decade ago.

In previous occasions also, they, as a group or as few individuals from the same batch, contributed “Little grains of sand” to make the “Mighty” building.

This time discussion was how we could have the guest rooms of Kanchenjunga more comfortable to stay. Nowadays air conditioners are more of essential items at hotel / guest houses rather than luxury. This Convention Centre holds prestigious seminars, eminent speakers / guests do come and stay here. This further justifies that guest rooms should have minimum comfort of air conditioning.

Since expenditures at Project Kanchenjunga are very carefully prioritized based on the available funds, the central project team could not even think of providing air conditioners at guest rooms. We deliberated and after lot of brainstorming, decided to bail them out by sponsoring 5 nos. ACs for the guest rooms.

Next morning, we conveyed this proposal to Governing Body of The JGEC Alumni Association that Batch 1991 is willing to sponsor 5 nos. ACs for the guest rooms at Kanchenjunga. It was merrily accepted at the GB and we were told to go ahead.

There comes the most challenging task. Our total budget was Rs.2 lacs. We appealed to all our group members to contribute. Within 15 days we reached the goal. Almost 40 members from us contributed. At times, it was so exciting and enthusiastic that we had to return excess money from one of our members.

However, no project can run without any hiccups. After we raised the above fund, we came to know that building's electrical load, wiring, phase, etc. are not adequate to cater to the requirement of 5 nos. ACs. The whole enthusiasm of our group was fused at that moment. We were clueless with what to do with Rs.2 lacs raised from 40 members. Lot of brick batting and criticisms started following among the group members. We came up with a solution that we handover the money to The JGEC AA and let them open a Fixed Deposit at bank of same amount. Once build infrastructure gets ready, they will liquidate that to procure those 5 nos. ACs. Governing Body agreed and FD was opened.

During January 2026, we were told that building infrastructure was almost ready and that we could start procurement action.

Finally in February 2026, the project team installed 5 nos. ACs (1.5TR each) in the guest rooms. With the fund we raised, we could even manage to supply dedicated voltage stabilizer for each AC and cost provision for next 5 years' AMC charges has been made.

We, 1991 pass out batch, again take pride in directly participating in the making of Kanchenjunga and giving back to our Alma Mater.

Tensors without Tension

Abhijit Sarkar (ME 2000)



The author is Professor & Head of the Machine Design Section, Department of Mechanical Engineering IIT Madras. He is a life-long student in diverse areas of Mathematics & Classical Mechanics. He has been an unapologetic bookworm/ghoshu/theoretician in his career. A naturally-stupid (not Artificially intelligent) old school prof, he is a passionate practitioner of the chalk-and-talk pedagogy.

Introduction

Physical quantities are classified as scalars, vectors and tensors. While the former two are well-understood amongst the student community, tensors have been disreputed as being a preposterous, esoteric concept conjured by some cranky, nutty professors only to inflict savage misery to the students on the night before the examination.

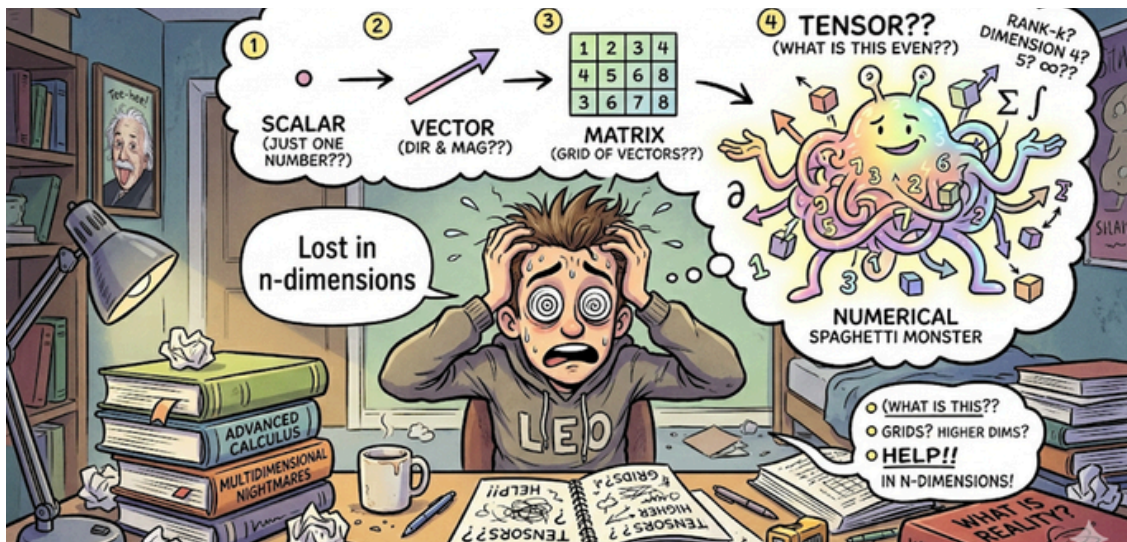


Figure 1: God created Integers. Which devil created tensors? (Image generated by Gemini AI).

The author had his share of frustration in climbing a steep learning curve. He did fall from the cliff on multiple occasions. But with the able guidance of his teachers, he was able to dust-off the initial frustration. It's payback time! The objective of writing this article is to hand-hold the reader and initiate him/her into the beautiful world of Mathematical Physics. So hold my hands, I will be your guide. Its going to be a difficult & challenging trek for sure. But as is true with any difficult trek, the view is breathtaking. Hopefully, you will have your AHA moments as you chug along!



Figure 2: Dar ke age jeet hai (Image generated by Gemini AI).

Scalars: The journey starts here

Scalars are represented by numbers, real numbers (mostly, but not always). While this definition may seem obvious to the reader, the emphasis on the word representation cannot be overstated. The actual physical quantity (say, temperature, pressure, charge, etc.) is abstract. We are representing the physical quantity as a real number. This is similar to answering the philosophical question “Who am I?” The naive answer seems to be “I am Coach Golu” - but that is my name. Indeed, in the hostel slang, I may/will have a different name (which is unknown to me). But for the purpose of this article, I am represented as “Coach Golu”. Similarly, temperature is not just a real number. It is actually difficult to describe what temperature (or for that matter any physical quantity) is. However, we completely understand the statement that the ambient temperature is 300C. This implies that a certain thermometer displayed 300C as the measurement of the temperature. A moment of thought should convince the reader that this measurement process involves several man-made (apologies for the gender bias) conventions and assumptions.

In dealing with scalars, the good news is that this representation is independent of the observer. In other words, it does not matter whether the observer is standing, sleeping or running. The temperature of a point is the same number. Scalars (such as temperature) are therefore observer-invariant. Figure 3 illustrates this aspect. This seemingly innocuous feature of observer-invariance will go for a toss in the case of vectors.

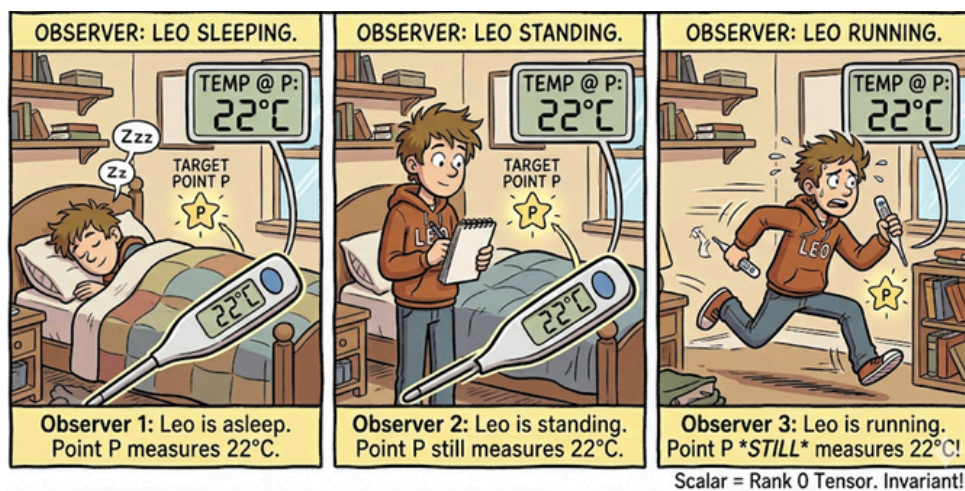


Figure 3: Scalars are observer invariant (Image generated by Gemini AI).

Vectors: Level-up the game

At an abstract level, a vector is an arrow. A vector quantity (such as force, velocity, heat flux, etc.) has both magnitude and direction. Magnitude can be represented as a positive number. However, the representation of the direction is non-trivial. A naive idea would be to hand-wave or perhaps to actually draw the vector as an arrow using some scale to represent the magnitude. This could possibly be done for two-dimensional vectors. However, extending this idea to three-dimensional space would pose serious challenges. Indeed, drawing three-dimensional arrows on the two-dimensional plane of paper using the front view / top view would take us on a digressive path of engineering drawing. Further, this path would have a dead end at higher dimensions (yeah, there are higher dimensions, but not in the scope of this article). We need a better mathematical representation.

So here is the recipe. We endow each observer (an earthling, not an alien) with three scales with the following attributes: -

- (1) All three scales are affixed and glued to the head of the observer.
- (2) The first scale points from left to right.
- (3) The second scale points along the direction of his/her nose
- (4) The third scale points along the direction from his/her feet to the head.
- (5) All the three scales are mutually perpendicular.

This arrangement is illustrated in Figure 4. Technically, the three scales constitute the basis vector for the observer. Given any vector (rather arrow), we drop the projection (shadow) of the vector along these three perpendicular scales. Let the length of the projections along the three directions be v_1 , v_2 , v_3 , respectively. For the purpose of book-keeping, the observer chooses to represent the geometric object (vector/ arrow) as a columnar array in the following manner: -

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

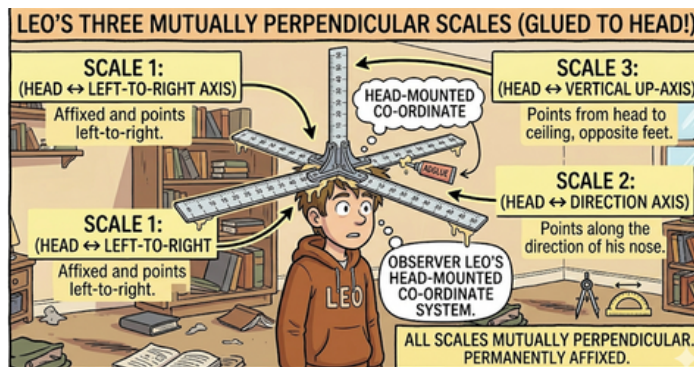


Figure 4: A coordinate axis (or basis) affixed to the observer (Image generated by Gemini AI).

This is illustrated in Figure 5. It must be emphasised that the above columnar array is a representation or an encoding of the actual geometric arrow. The reader should be agreeable to this recipe of the representation. Definitely, it is more natural than representing humans by their names. The same person may have different names and conversely different persons do have the same name. Unlike humans, with every arrow, there is a unique columnar array and vice versa. The representation of the arrow in this manner naturally depends on the orientation of the head of the observer (which in turn depends on the chemical composition of the liquids (s)he consumed).

The fact that the representation of the vector is observer-dependent can be a major source of confusion. Two observers (one standing and the other sleeping) will have their scales oriented in a different fashion. Thus, their book-keeping description of the abstract geometric vector as a columnar array (mathematically a column matrix) will naturally differ. Thus, the numbers v_1 , v_2 , v_3 are strictly observer dependent and not an intrinsic property of the geometric object viz. The vector. In other words, the components do not have any physical significance.

Despite the difference between the components reported by the two observers, there are certain invariant quantities about the vectors which the two different observers agree on. For example, the (Euclidean) length of the vector given by is identical (refer Figure 6 for an illustration). Similarly, the angle between any two vectors (whatever be its formula) is calculated to be the same by both observers. Thus, these quantities are intrinsic properties of the vector and independent of the observer. It is natural that the laws of physics are observer-independent and should thus depend only on the intrinsic observer-invariant quantities. This very profound idea lies at the heart of mechanics, elasticity, constitutive modeling, electromagnetics, and finally relativity.

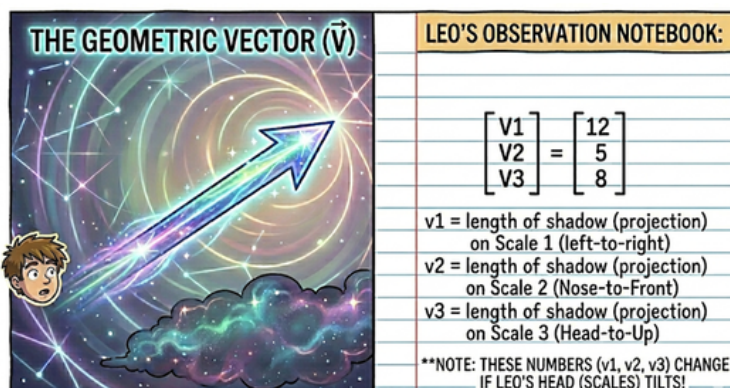


Figure 5: The geometrically abstract vector is represented as a columnar array or column matrix (image generated using Gemini AI).

There are two most important operations that we need to note carefully at this stage. These are vector addition and scalar-vector multiplication. The scalar-vector multiplication stretches the vector/arrow (in case the scalar is a positive number greater than one). If the scalar involved is between 0 and 1, the vector/arrow is shortened. In both these cases, the direction of the vector/ arrow is unchanged. Finally, if the scalar is negative, then the arrow flips its direction along with a stretch (for scalars less than -1) or shortening (for scalars between -1 and 0) of its length. In tandem with the geometric picture the notebook entries of the vector components get multiplied with the scalar in question. This is illustrated in figure 7.

The vector addition on the other hand, combines two vectors to produce another resultant vector. Geometrically, the operation is easy to construct in a two-dimensional plane. Using the two vectors (red and blue in Figure 8), we draw a parallelogram. The diagonal of the parallelogram is the resultant addition of the vectors (shown in green in Figure 8). The same idea can be extended in the three-dimensional (or higher dimensional) space. Note, that two arbitrary vectors (arrows) in three dimensional space lie in a plane. The action of vector addition of the two vectors happens in the plane containing these two vectors. By repeating this operation recursively, multiple vectors can be added. The vector addition operation can equivalently be represented using the columnar array representation by adding each entry of the array. This applied to two-dimensional plane, three-dimensional space as well as any higher dimensional hyperspace (you will go there for sure but not right now). Figure 8 presents a schematic illustration of the vector addition operation.

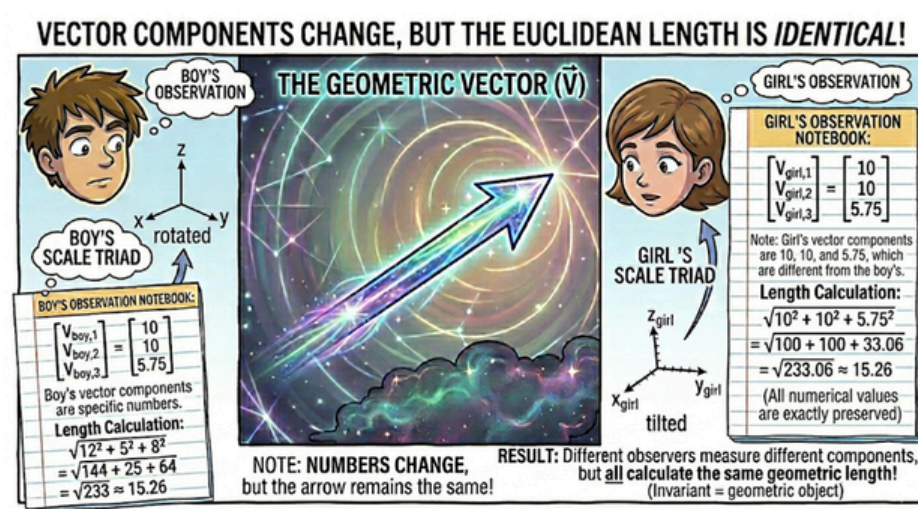


Figure 6: The representation of the vector/ arrow in component form is observer dependent but the magnitude is independent of the observer (image generated using AI).

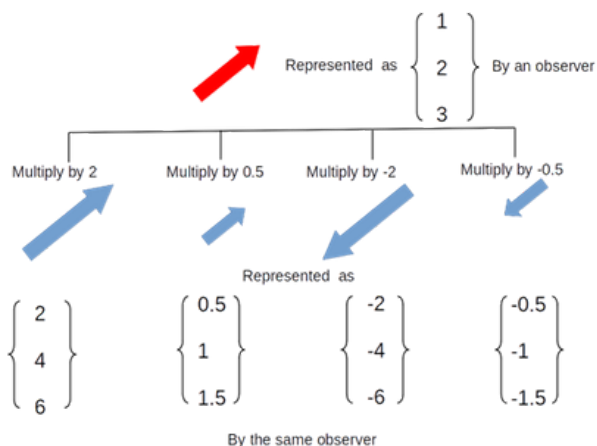


Figure 7: Schematic illustration of scalar-vector multiplication. The geometric operation of stretching, shortening or flipping the direction is represented by multiplication of the columnar entries in the array.

The operations of scalar-vector multiplication and vector-vector addition lie at the heart of encoding the vector in columnar array form. Any vector can be decomposed into its constituent elements which are projection on the axes (also called basis vectors). These constituent vectors can be added to get back the vector. This concept is schematically illustrated in Figure 9. This act of decomposing an arbitrary vector into its constituent elements is called superposition. Each observer can carry an arbitrarily oriented basis vector on his/her head. Thus, the numbers representing the vector (viz. x, y, z) in the illustration can differ between observers.

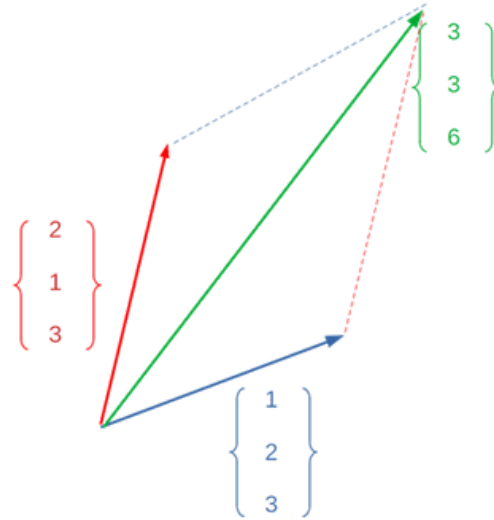


Figure 8: Schematic representation of the vector addition operation. The red and the blue vectors are added, resulting in the green vector. Their columnar array representation is shown using the same color code.

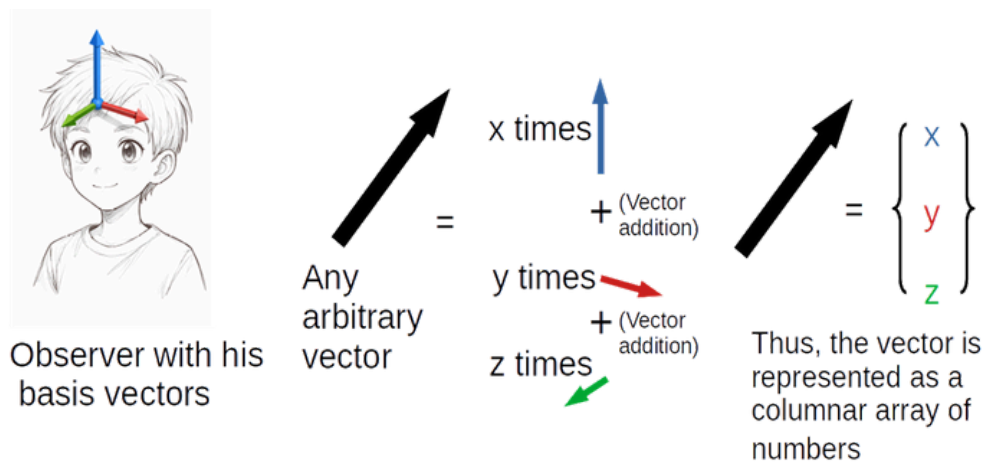


Figure 9: Given any arbitrary vector, an observer can decompose the vector into its constituent projections along the basis vectors.

Note, as per the convention illustrated in Figure 9, the blue basis vectors will have column matrix representation as $\{1, 0, 0\}$, the red basis vector will have a column matrix representation as $\{0, 1, 0\}$ and the green vector will have a column matrix representation as $\{0, 0, 1\}$.

Tensor: A linear transformation

In line with the style of this article, we first define a tensor in an abstract fashion. A tensor is an object which takes an input vector and transforms it into an output vector. Figure 10 illustrates the idea. So in the abstract setting, the vector is an arrow and the tensor is a machine which takes an input vector/ arrow and gives out another vector/arrow as output. The abstract transformation in the current context is required to abide by certain rules. Specifically, the transformation rules must respect the operations of vector addition and scalar-vector multiplication that we have defined earlier for vectors.

The input vector indicated by a solid arrow transforms to the vector indicated by a dashed arrow of the same color as indicated in Figure 11. The addition of the red and blue vectors (solid and dashed, respectively) results in the green vector (solid and dashed, respectively). One could add the two vectors and then transform it through the tensor machine, get their individual outputs and then add the output vectors. In either of these cases, the resultant is the same. In other words, the operation of vector addition followed by tensor transformation is the same as tensor transformation followed by addition. Equivalently, the order of vector-addition operation and tensor transformation operation can be interchanged.

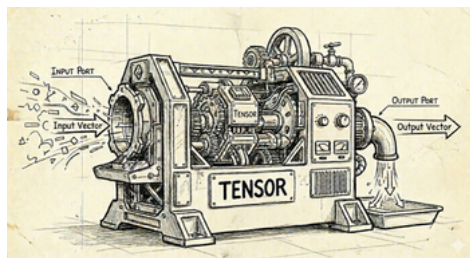


Figure 10: A tensor (or a Linear transformation) is machine taking one input vector and transforming it to another output vector. The vectors are arrows (in the abstract representation) but to an observer it is a columnar array. Image created by Gemini AI.

Similarly, the scalar-vector multiplication operation needs to be preserved by the tensor transformation operation. In particular, if the input vector is scalar multiplied by a certain factor then the output vector is also scalar multiplied by the same factor. This is illustrated in Figure 12.

A transformation preserving vector-addition and scalar-vector multiplication is called a linear transformation. Thus, a tensor is specifically a linear transformation.

Having described the abstract definition of tensor, it is apt that we have some concrete examples at this point. It should convince the reader that geometric operations such as rotation and reflections are linear transformations preserving the rules of vector-addition and scalar multiplication. The Moment of Inertia tensor is a physical quantity that transforms the angular velocity vector into the angular momentum vector. The stress tensor transforms the unit normal vector to the traction vector. The traction vector is the force vector divided by the magnitude of the area. Although, electric field and magnetic field are individually vectors, together they form an electromagnetic field tensor in the four-dimensional spacetime.

Now that we have the abstract definition and a few concrete examples of tensors, the final question (addressed in this article) is how do we represent them? We had a nice encoding of the vector which in abstract fashion was an arrow but to the observer was a set of three numbers arranged in a columnar fashion. The reader should have been convinced about the efficacy of such a book-keeping. It turns out that the tensor can be represented as a two-dimensional square array or matrix. The recipe for conjuring this matrix representation of the tensor is described in the final part of the article.

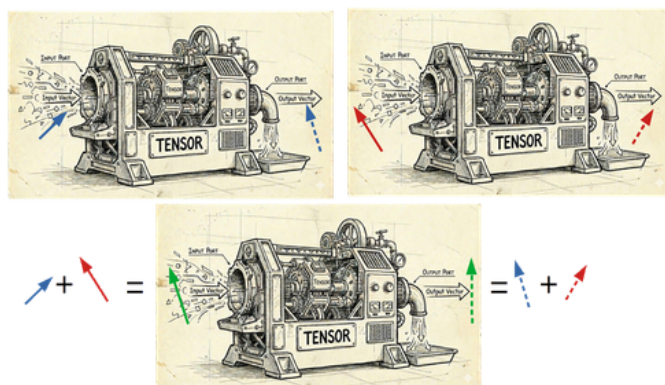


Figure 11: Schematic illustration of Tensor preserving the law of vector addition. Tensor transforms an input vector (indicated by solid arrow) to another vector (indicated by dashed arrow of the same color). If we add the input vectors and then transform it through the tensor, then the resultant vector is the same as the addition of the two output vectors.

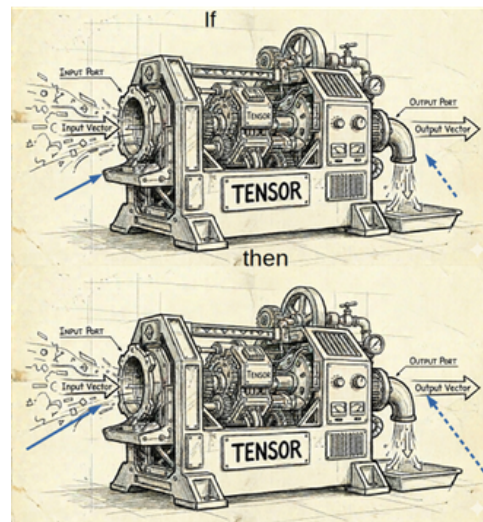


Figure 12: Schematic illustration of Tensor preserving the scalar-vector multiplication. If the input vector (indicated as solid arrow) is scaled, then the output vector (indicated as dashed arrow) is also scaled by the same factor.

Matrix representation of Tensor

In order to represent the tensor, we need to ensure that we have a rule by which we can get the output vector for any input vector. Except in simple cases (such as the tensor representing a scaling operation), this can be a herculean task. A divide-and-rule approach can be employed to make life simple. Using this approach, it is enough for the observer to know how the tensor machine performs on each of the vectors glued to his/her forehead. Let me explain.

Assume the observer knows the action of the tensor on each of the basis vectors glued to his/her head. In particular, the red/green/blue solid vector transforms to the red/green/blue dashed vector. The red/green/blue dashed vectors each have their column matrix representation. This concept is illustrated in the schematic shown in Figure 13.

Explicitly, the blue vector, represented as

$$\begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix} \text{ is transformed to } \begin{Bmatrix} T_{11} \\ T_{21} \\ T_{31} \end{Bmatrix}$$

Similarly, the red vector, represented as

$$\begin{Bmatrix} 0 \\ 1 \\ 0 \end{Bmatrix} \text{ is transformed as } \begin{Bmatrix} T_{12} \\ T_{22} \\ T_{32} \end{Bmatrix}$$

Finally, the green vector is represented as

$$\begin{Bmatrix} 0 \\ 0 \\ 1 \end{Bmatrix} \text{ is transformed as } \begin{Bmatrix} T_{13} \\ T_{23} \\ T_{33} \end{Bmatrix}$$

Again, for book-keeping purposes, the observer stores the collection of the output vectors in a two-dimensional array (square matrix if you like) as follows: -

$$\begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix}$$

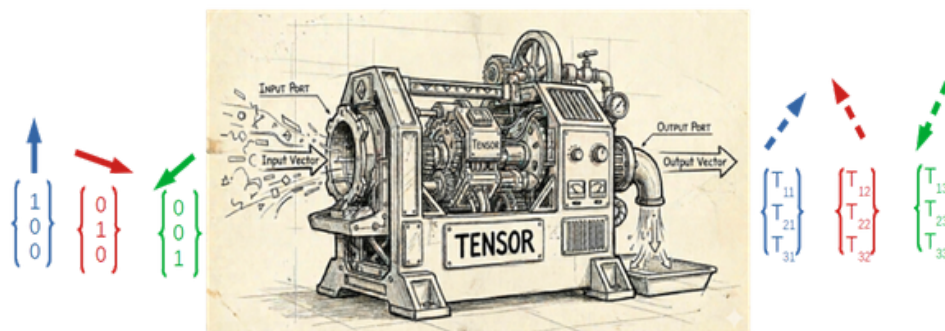


Figure 13: Schematic illustration showing the tensor transformation on the basis vectors (shown as solid arrows on the input side). The output for each of these basis vectors is indicated with the same colour but with dashed arrows. The column matrix representation of these vectors is shown adjacent to the arrows.

All the information the observer needs to know about the tensor in order to determine the output vector for any input vector is contained in the above two-dimensional array. This is accomplished in the following manner. The input vector is decomposed as a superposition of multiples of the basis vectors (as was done in Figure 9). Exploiting the property of linear transformation being respectful of the scalar-multiplication and vector addition, each constituent element of the decomposition is transformed and then added up; giving the transformed output vector. This operation is illustrated in Figure 14.

Thus, the reader should be convinced that the tensor operation can be completely represented by the two-dimensional array or matrix as given above. The matrix has just adequate (and no surplus or redundant) information in order to enable the observer to determine the output vector for any input vector.

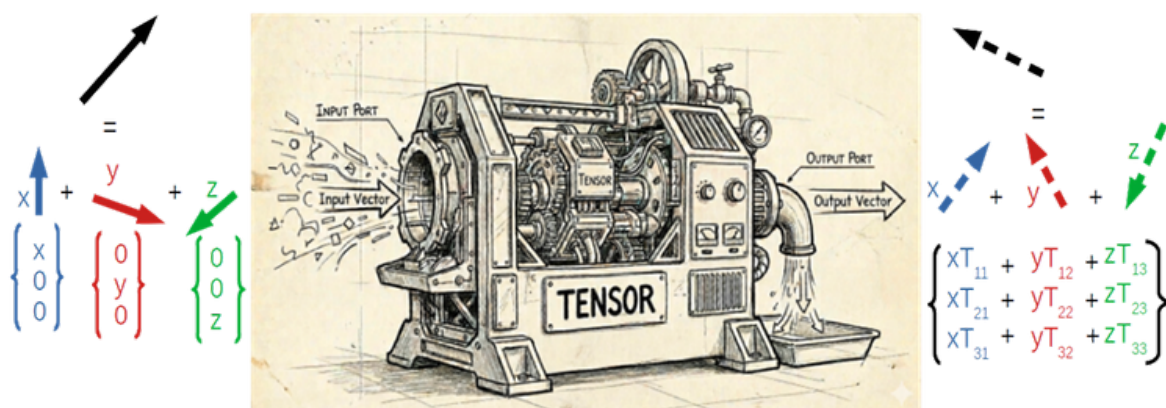


Figure 14: In order to know how an arbitrary vector (shown as black solid arrow) transforms due to the tensor operation, one needs to decompose the vector into its constituent elements along the basis. Each constituent element is transformed and then superposed to get the resultant output vector (shown as black dashed arrow).

This may seem strange. Since the matrix itself was constructed by knowing the action of the tensor on just three basis vectors glued at the head of the observer. Yet, this all-powerful matrix is able to conjure up the action of the machine on any vector. This is made possible due to the requirements of the transformation being linear.

Indeed, vectors are represented as column matrices, and tensors are represented as square matrices by an observer by using his/her basis vectors (still glued to his/her head). Thus, the abstract cosmic vector-arrow and the rustic tensor-machine are boiled down to a set of numbers, arranged in specific patterns. The operation of obtaining the output vector from the given input vector is precisely the matrix-vector multiplication operation. Just like the columnar array representation of the vector was dependent on the triad or basis vectors (glued to the head of the observer), the two-dimensional array or square matrix representation of the tensor is dependent on the observer's basis.

The abstract cosmic-arrow is the vector. The choice of basis enables the observer to represent the same as a set of numbers (arranged in columnar fashion). Similarly, the abstract rustic machine is the tensor. The choice of basis enables the observer to represent the same as a set of numbers, this time arranged in a two-dimensional square form. With a different observer, the representation will differ, but the abstract object will remain the same. After all, what's there in a name or a representation?

Laws of physics stated in tensorial form will be independent of the observer. Thus, the numbers (technically components) representing the vectors or tensors don't matter. However, from these numbers we can concoct some observer-invariant quantities. We have noted earlier that the length of the vector given by the formula is one such invariant quantity. Similarly, for tensors, there are various invariant quantities which can be derived from the component representation but are observer independent. Some examples of such invariant quantities are trace (sum of diagonal entries of the matrix), determinant, eigenvalues, etc. Thus, although the matrix entries representing a tensor differ between the observers, the above invariant quantities are calculated as identical by all observers. All observers have finally reached Rome !

Matrix representation of Tensor

All good (and bad) things must come to an end. This may be a good place to pause and soak in the view. We have just reached the base camp of our climb. There is miles to go before we get to the peak of this Tensor-verse. Hopefully, the readers will take home the message that scalars are represented as numbers in an observer-independent fashion, vectors are represented as column matrices, and tensors are represented as square matrices. The representation of vectors and tensors depends on the observer. Each representation is a human attempt to put down in black and white the nirakar and the abstract quantity. On that philosophical note, let us stop here. Thank you for joining me in this journey. It has been my pleasure to take you through this guided tour. In case you are interested in learning more about this topic, I am leaving you with some references. I do intend to update this material as and when I get the opportunity. The updated material will be posted on my website (<https://sites.google.com/site/abhijitmeiitm/>).

Acknowledgement

The author thanks The Jalpaiguri Government Engineering College Alumni Association for giving him an opportunity to put up this article. During preparation of this article, the author was assisted with AI tools (Gemini and Chatgpt).

References:

- The theoretical minimum series by Leonard Suskind and accompanying videos available on the internet. It's a 4 volume series. Volume 1 & 3 are my favourites. The book and the lecture series is an informal, popular science lecture series.
- Vector & Tensor Analysis by Borishenko and Tarapov. This is a classic book and is now available in Dover Edition.
- The very basics of tensor by Neils K Oijord. This is a formal book and is not meant for the faint-hearted.

Flashbacks



Kanchenjunga Inauguration
dtd. 22.08.2022



Alumni Association Initiative: Readers' Corner

Spoken English Class under umbrella of Mentorship Program



Alumni Spotlight

"Success is the sum of small efforts, repeated day in and day out" - Robert Collier



Remaining humble - Giving back, spreading happiness!

Pay a visit to Kanchenjunga to see yourself!



KANCHENJUNGA- PROUD POSSESSION OF THE ALUMNI

Let's Keep "Kanchenjunga" Thriving! - Appeal for Annual Contribution.



Dear Fellow Alumni,

Thanks to the incredible dedication and support from all of you, our dream project, "Kanchenjunga," is almost complete! This is a huge achievement for our community, but it's also just the beginning. All of us is willing to see our very own Kanchenjunga as beautiful and glamorous in the coming years also as it is now.

To ensure "Kanchenjunga" remains a valuable asset for both students and alumni for a long time, we need to set up a solid plan for its regular operation and maintenance. We have thought that the time has come to engage one personnel, who will be responsible for daily operation and regular maintenance of the Kanchenjunga.

The estimated monthly expenditure in this head will be approximately 40,000/-. After a detailed discussion at our last Annual General Meeting, it was agreed upon that a subscription-based support system of the Alumni may be envisaged which is expected to be of great help, alongside other existing funding methods like financial support and one time donations.

Therefore, each and every member of the big Jolu fraternity are requested to come forward and commit to an annual contribution of at least ₹1000/-. If more and more alumni come forward, it will help in build up a coffer for the Alumni activities.

We've already had a fantastic start—at our recent AGM, 24 members generously contributed ₹24,000 to get the initial impetus.

I, on behalf of Alumni Association personally appeal to you to support this effort. Please scan the QR code below and contribute ₹1000/- for one year.

Your support means the world to us.

Best regards,
Secretary,
JGEC AA



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Thank You